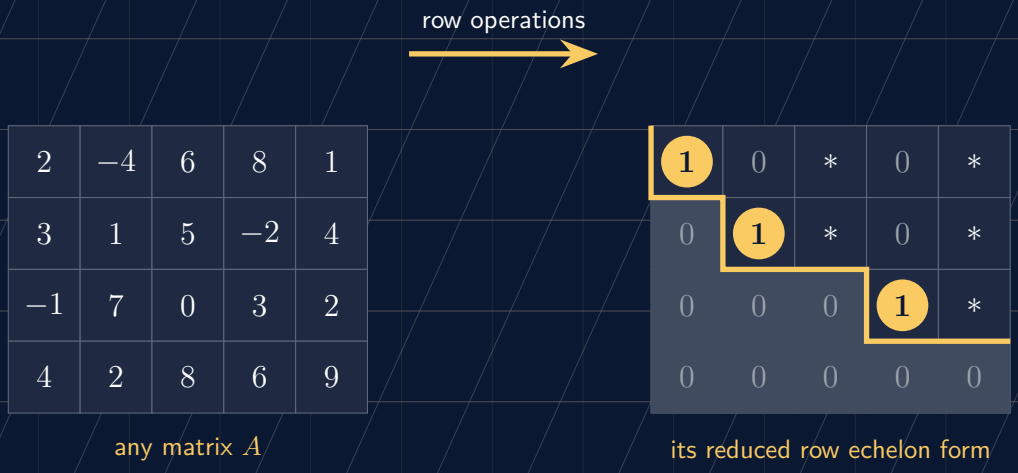
LINEAR ALGEBRA · LECTURE 2

Gaussian & Gauss–Jordan Elimination

echelon form and reduced echelon form

Augmented matrices · Leading 1's · The staircase · Lines and planes
One / none / infinitely many solutions · Free variables · Homogeneous systems · Pivots



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Lecture notes for BS students | every row operation written out, many pictures, a summary sheet

Built on: Anton & Rorres, *Elementary Linear Algebra* (§1.2 Gaussian Elimination) · G. Williams, *Linear Algebra with Applications* (§1.2 Gauss–Jordan Elimination)

Contents

1 The story: why do we put a system into a matrix? 1

1.1 Solving with equations: we keep rewriting the variables . . . 1

1.2 The augmented matrix . . . 2

1.3 The three elementary row operations . . 2

1.4 The same system with the matrix 3

2 Reduced row echelon form: the heart of linear algebra 3

3 Echelon forms: the definitions 5

3.1 Leading entry and leading 1 . . . 5

3.2 Row echelon form and reduced row echelon form . . . 5

3.3 The staircase picture . . . 6

4 Using the definition: which form is it? 6

4.1 A four-question test . . . 6

4.2 A gallery with staircases . . . 7

5 What row operations do to the pictures 8

5.1 Two lines: the lines move, the meeting point does not . . . 8

5.2 Three planes: the same story in space . 8

6 Gaussian and Gauss–Jordan elimination 9

6.1 The algorithm (Anton's six steps) 9

6.2 Anton's example, one step at a time . . 10

6.3 Williams' version: column by column . . 11

6.4 Gaussian elimination and back-substitution . . . 11

7 One solution, no solution, or infinitely many 14

7.1 Reading an echelon form . . . 14

7.2 In pictures . . . 14

7.3 Free variables: which ones to choose, and which never . . . 19

8 Homogeneous systems 20

8.1 Trivial and nontrivial solutions 20

8.2 Free variables in homogeneous systems . 21

9 The important theorems in one place 24

10 Some facts about echelon forms; pivots 24

11 Students' common misconceptions 25

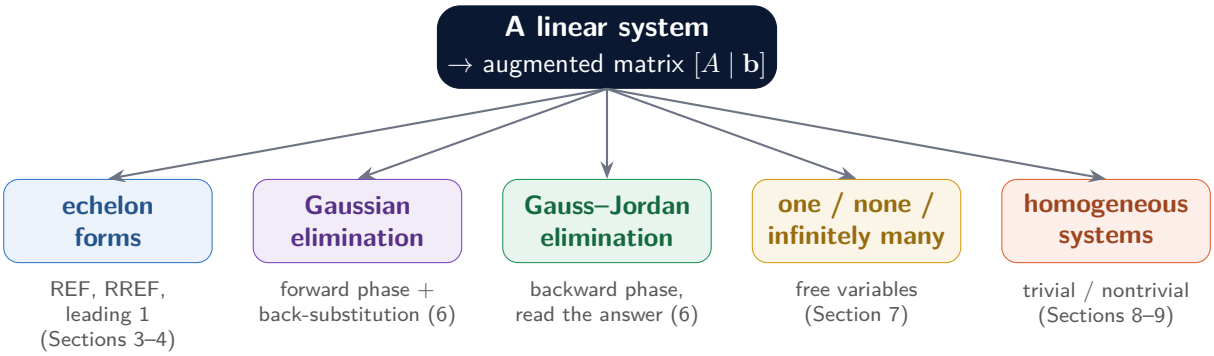
12 Solved problems on echelon and reduced echelon forms 28

13 Everything on one page 32

14 Check yourself 33

How to read these notes

- Every row operation is written **on its arrow**: $\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ \mathbf{0} & \mathbf{-3} & \mathbf{-6} \end{array}\right]$.
- The row that has just **changed** is printed in **orange**. In a final echelon matrix every leading 1 is circled: (1). Nothing is skipped: every intermediate matrix is shown.
- Notation: R_i = row i. $R_i \leftrightarrow R_j$ = swap rows i and j. $R_i \to cR_i$ = multiply row i by $c \neq 0$. $R_i \to R_i + cR_j$ = add c times row j to row i. Operations that use the same pivot row are stacked on one arrow.
- Boxes: Definition | In simple words | Theorem | Example | Common mistake | Where is this used?
- Tags such as [Anton §1.2 · Ex. Set 1.2, No. 5] tell you where a problem comes from in the textbook. All the numbers in these notes were checked by computer (exact fractions).



1 The story: why do we put a system into a matrix?

1.1 Solving with equations: we keep rewriting the variables

Take the system [Anton §1.2 · Ex. Set 1.2, No. 5]

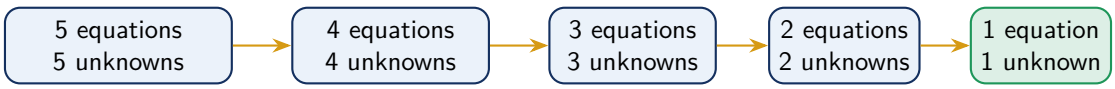
$$\begin{cases} x_1 + x_2 + 2x_3 = 8 \\ -x_1 - 2x_2 + 3x_3 = 1 \\ 3x_1 - 7x_2 + 4x_3 = 10 \end{cases}$$

At school we eliminate: add equation (1) to (2), subtract 3×(1) from (3), and so on. Look at what we actually write on the page:

$$\begin{aligned} x_1 + x_2 + 2x_3 &= 8 \\ -x_1 - 2x_2 + 3x_3 &= 1 \\ 3x_1 - 7x_2 + 4x_3 &= 10 \end{aligned} \xrightarrow[(3)-3(1)]{(2)+(1)} \begin{aligned} x_1 + x_2 + 2x_3 &= 8 \\ -x_2 + 5x_3 &= 9 \\ -10x_2 - 2x_3 &= -14 \end{aligned} \xrightarrow{(3)-10(2)} \begin{aligned} x_1 + x_2 + 2x_3 &= 8 \\ -x_2 + 5x_3 &= 9 \\ -52x_3 &= -104 \end{aligned}$$

the symbols $x_1, x_2, x_3, +, =$ are copied again and again: they never change

Every time, the **letters** x_1, x_2, x_3, the plus signs and the equals signs are copied again. They carry **no information**: only the **numbers** change. Now imagine 5 equations in 5 unknowns:



at every arrow all remaining equations, with all their letters, are written out again

That is a lot of copying, and every copy is a new chance to drop a sign or a variable. So we make one simple decision:

The decision

Do not rewrite the variables. Keep only the numbers. Write the coefficients in a rectangle, put the constants in an extra column after a bar, and work only with the **rows** of this rectangle. This rectangle is the **augmented matrix** of the system.

1.2 The augmented matrix

$$\begin{cases} 1x_1 + 1x_2 + 2x_3 = 8 \\ -1x_1 - 2x_2 + 3x_3 = 1 \\ 3x_1 - 7x_2 + 4x_3 = 10 \end{cases} \xrightarrow{\text{keep the numbers}} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right]$$

constants **b** (right of the bar)

row i of the matrix = equation i

coefficient matrix A (column j belongs to x_j)

Definition 1 (Augmented matrix)

For a system of m linear equations in n unknowns, the **augmented matrix** $[A \mid \mathbf{b}]$ is the $m \times (n+1)$ matrix whose row i lists the coefficients of equation i (in the order $x_1, \ldots, x_n$) followed by its constant term.

In simple words

Equation $\longleftrightarrow$ row. Variable $x_j \longleftrightarrow$ column j. The bar is the "=" sign. A missing variable is a 0 in its column (never just leave a gap).

1.3 The three elementary row operations

What we did with equations becomes three moves on the rows:

Type	On equations	On rows	Example
1. Swap	interchange two equations	$R_i \leftrightarrow R_j$	$\left[\begin{array}{cc\|c} 0 & 2 & 4 \\ 1 & 1 & 3 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc\|c} \mathbf{1} & \mathbf{1} & \mathbf{3} \\ \mathbf{0} & \mathbf{2} & \mathbf{4} \end{array}\right]$
2. Scale	multiply an equation by $c \neq 0$	$R_i \to cR_i$	$\left[\begin{array}{cc\|c} 1 & 1 & 3 \\ 0 & 2 & 4 \end{array}\right] \xrightarrow{R_2 \to \frac{1}{2}R_2} \left[\begin{array}{cc\|c} 1 & 1 & 3 \\ \mathbf{0} & \mathbf{1} & \mathbf{2} \end{array}\right]$
3. Add	add $c\times$ one equation to another	$R_i \to R_i + cR_j$	$\left[\begin{array}{cc\|c} 1 & 1 & 3 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{cc\|c} \mathbf{1} & \mathbf{0} & \mathbf{1} \\ 0 & 1 & 2 \end{array}\right]$

The last matrix says $x_1 = 1$, $x_2 = 2$. Each move can be undone by a move of the same type, so **the solution set never changes** (Section 5 shows this with pictures).

1.4 The same system with the matrix

Now solve the system above using only numbers. The steps are *exactly* the steps we did with the equations, but nothing is copied in vain:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ \mathbf{0} & \mathbf{-1} & \mathbf{5} & \mathbf{9} \\ \mathbf{0} & \mathbf{-10} & \mathbf{-2} & \mathbf{-14} \end{array}\right]$$

$$\xrightarrow{R_2 \to -R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ \mathbf{0} & \mathbf{1} & \mathbf{-5} & \mathbf{-9} \\ 0 & -10 & -2 & -14 \end{array}\right] \xrightarrow{R_3 \to R_3 + 10R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ \mathbf{0} & \mathbf{0} & \mathbf{-52} & \mathbf{-104} \end{array}\right]$$

$$\xrightarrow{R_3 \to -\frac{1}{52}R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{2} \end{array}\right] \xrightarrow[R_2 \to R_2 + 5R_3]{R_1 \to R_1 - 2R_3} \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{4} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{1} \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & 3 \\ 0 & \boxed{1} & 0 & 1 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

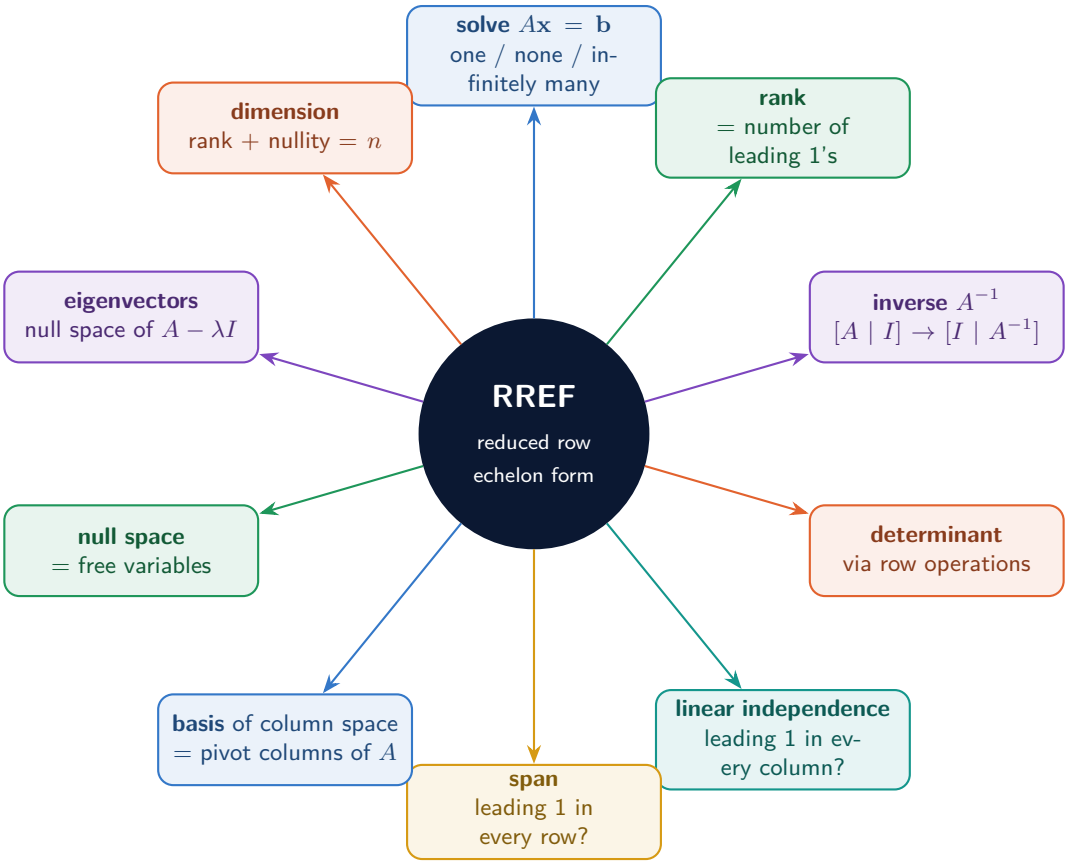
The last matrix is the system $x_1 = 3$, $x_2 = 1$, $x_3 = 2$, which we read off **without any back-substitution**. This final shape is the hero of this lecture: the **reduced row echelon form**.

Where is this used?

Gauss used this method around 1801 to compute the orbit of the asteroid Ceres from a few observations; it reappeared a year later almost exactly where he predicted. Wilhelm Jordan (1888) popularised the "reduce all the way" version. Today every computer program that solves linear systems (MATLAB, NumPy, Excel's solver, engineering software with millions of unknowns) is built on these row operations. [Anton §1.2 · Historical note]

2 Reduced row echelon form: the heart of linear algebra

Before we define anything, here is why this lecture matters more than almost any other. In the rest of the course, a surprisingly large number of questions are answered in **the same way**: *row reduce, then look at the leading 1's.*



The one idea to take away

Row operations **do not change** the solutions of a system, the dependencies among the columns, or the rank of a matrix; they only make these things **visible**. The reduced row echelon form is the clearest possible picture of a matrix, and it is **unique**: whatever sequence of operations you use, you always reach the same RREF. That is why almost every later chapter starts with "reduce

the matrix to RREF".

Question in a later chapter	What you look at in the RREF
Is $A\mathbf{x} = \mathbf{b}$ consistent? How many solutions?	a row $[0 \cdots 0 \mid c]$, $c \neq 0$? free variables? (this lecture)
What is the rank of A?	the number of leading 1's
Is A invertible?	is $\mathrm{rref}(A) = I$? (and $[A \mid I] \to [I \mid A^{-1}]$)
Are the columns of A independent?	is there a leading 1 in *every column*?
Do the columns span $\mathbb{R}^m$?	is there a leading 1 in *every row*?
Basis of the column space / row space	pivot columns of A / nonzero rows of the RREF
Null space, its dimension (nullity)	solve $A\mathbf{x} = \mathbf{0}$: one vector per free variable
Eigenvectors for an eigenvalue λ	RREF of $A - \lambda I$, then its free variables

In simple words

Gaussian and Gauss–Jordan elimination are not "just a method for solving equations". They are the **microscope** of linear algebra. Learn them so well that you can do them half asleep: everything else in the course will be built on top of them.

3 Echelon forms: the definitions

3.1 Leading entry and leading 1

Definition 2 (Leading entry, leading 1)

In a row that is not all zeros, the **leading entry** is the *first nonzero number* when we read the row from left to right. If the leading entry is 1, it is called a **leading 1**. A row of zeros has no leading entry.



3.2 Row echelon form and reduced row echelon form

Definition 3 (Row echelon form, reduced row echelon form) [Anton §1.2; Williams §1.2]

A matrix is in **reduced row echelon form (RREF)** if it has these four properties:

(1) If a row is not entirely zeros, its first nonzero number is a 1 (a **leading 1**).

(2) Rows consisting entirely of zeros are grouped together at the **bottom**.

(3) In any two successive nonzero rows, the leading 1 in the lower row is **farther to the right** than the leading 1 in the higher row.

(4) Each column that contains a leading 1 has **zeros everywhere else** in that column.

A matrix with properties (1), (2), (3) (not necessarily (4)) is in **row echelon form (REF)**.

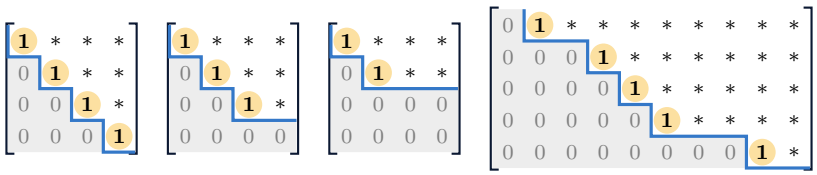
In simple words

REF = a **staircase**. Every step starts with a 1, each step is further right than the one above, below the staircase there are only zeros, and the empty rows sit at the bottom.
RREF = a **clean staircase**: in addition, each leading 1 is *alone* in its column (zeros above it as well as below). "Echelon" is the French word for a step of a ladder.
Every RREF is a REF; not every REF is an RREF. Williams calls the RREF simply the *reduced echelon form*.

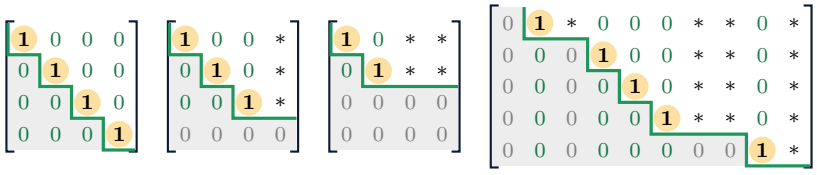
3.3 The staircase picture

Here $*$ means "any number" [Anton §1.2 · Example 2]. The shaded region under the staircase contains only zeros.

Row echelon form: zeros below the staircase, anything ($*$) above it



Reduced row echelon form: in addition, zeros above every leading 1



REF (forward phase stops here)
- zeros *below* every leading 1
- numbers above leading 1's allowed
- **not unique**: many REFs per matrix
- solve by **back-substitution**

RREF (backward phase done)
- zeros *below and above* every leading 1
- only free-column entries remain
- **unique**: one RREF per matrix
- read the solution **directly**

4 Using the definition: which form is it?

4.1 A four-question test

Check a matrix in this order

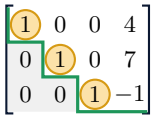
1. Circle the **leading entry** of every nonzero row. Is each one a 1? (property 1)
2. Are all zero rows at the bottom? (property 2)
3. Draw the staircase through the circles. Does it go **down and to the right** at every step? (property 3)

   If 1–3 hold: **REF**. Then:

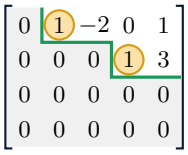
4. Look up and down each column of a leading 1. Is everything else 0? (property 4) Yes ⇒ **RREF**.

4.2 A gallery with staircases

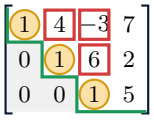
Green solid staircase = property (3) holds; red dashed = it fails. Gold circle = a correct leading 1; red circle = a leading entry that is not 1.



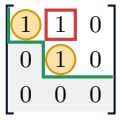
RREF (and REF)
clean staircase



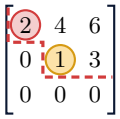
RREF (and REF)
steps may skip columns



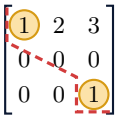
REF, not RREF
$4, -3, 6$ above leading 1's



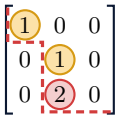
REF, not RREF
the 1 above the 2nd leading 1



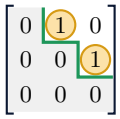
neither
leading entry $2 \neq 1$



neither
zero row not at the bottom



neither
row 3 steps back to the left

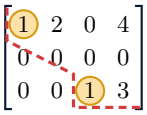


RREF (and REF)
first column may be empty

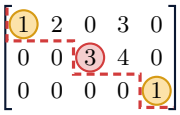
Red squares mark nonzero entries above a leading 1: allowed in a REF, forbidden in an RREF. [Anton §1.2 · Example 1; Ex. Set 1.2, No. 2]

Example Not reduced, and why [Williams §1.2 · after the definition]

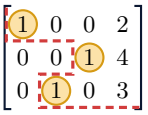
Each matrix below fails exactly one property, which is shown in red.



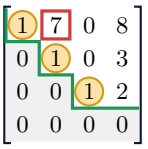
row of zeros not at the bottom: (2) fails



first nonzero element in row 2 is 3, not 1: (1) fails



leading 1 of row 3 is not to the right of the one in row 2: (3) fails



nonzero element (7) above the leading 1 of row 2: (4) fails

Example Reduced or not? (video) [Williams · Ex. Set 1.2, No. 1(c),(d),(f),(h)]

Legend: (1) = a correct leading 1, (2) = a leading entry that breaks a rule, 5 = a nonzero entry in the column of a leading 1 (allowed in REF, *not* in RREF).

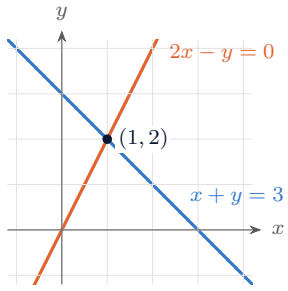
Matrix	Reduced?	Reason

(c)	$\begin{bmatrix} \boxed{1} & 2 & 5 & 6 \\ 0 & \boxed{1} & 3 & -7 \end{bmatrix}$	**no**	conditions (1)–(3) hold, so it *is* in row echelon form; but **(4)** fails: column 2 contains the leading 1 of row 2 but has the nonzero entry 2 in row 1.
(d)	$\begin{bmatrix} \boxed{1} & 4 & 0 & 5 \\ 0 & 0 & \boxed{2} & 9 \end{bmatrix}$	**no**	**(1)** fails: the first nonzero entry of row 2 is 2, not 1.
(f)	$\begin{bmatrix} \boxed{1} & 5 & 0 \\ 0 & 0 & \boxed{1} \\ 0 & 0 & 0 \end{bmatrix}$	**yes**	every nonzero row starts with a leading 1; the leading 1's step to the right; the zero rows are at the bottom; each column with a leading 1 has zeros everywhere else.
(h)	$\begin{bmatrix} \boxed{1} & 0 & 0 & 3 & 2 \\ 0 & \boxed{2} & 0 & 6 & 1 \\ 0 & 0 & \boxed{1} & 2 & 3 \end{bmatrix}$	**no**	**(1)** fails: the first nonzero entry of row 2 is 2, not 1.

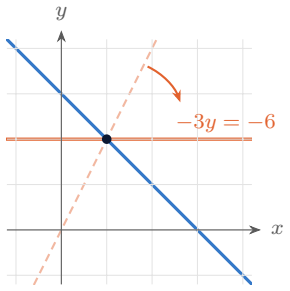
5 What row operations do to the pictures

5.1 Two lines: the lines move, the meeting point does not

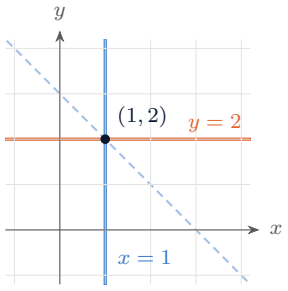
Each equation $ax + by = c$ is a line. Solving the system means finding the point where the lines meet. Watch the system $x + y = 3,\ 2x - y = 0$ during Gauss–Jordan elimination:



(a) original: $x + y = 3,\ 2x - y = 0$



(b) after $R_2 \to R_2 - 2R_1$: line 2 turns around $(1, 2)$ and becomes $-3y = -6$ (horizontal)



(c) after $R_2 \to -\frac{1}{3}R_2$ (same line, $y = 2$) and $R_1 \to R_1 - R_2$: line 1 becomes $x = 1$ (vertical)

The same steps on the augmented matrix:

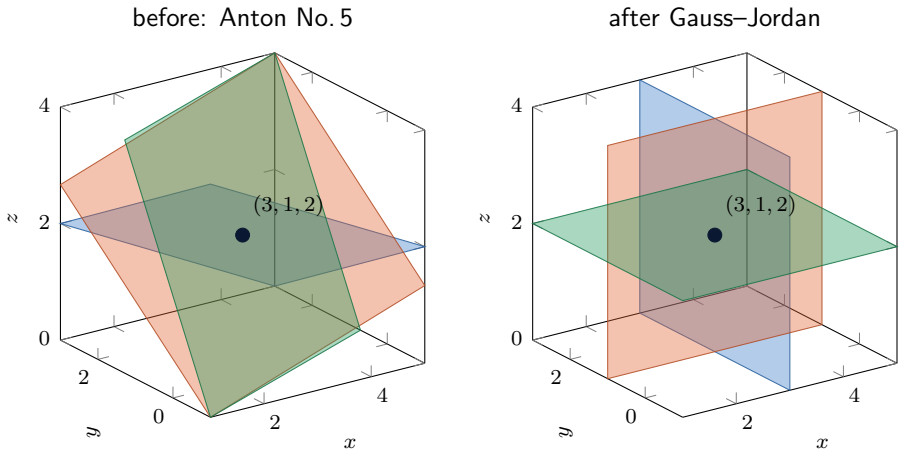
$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & -6 \end{array}\right] \xrightarrow{R_2 \to -\frac{1}{3}R_2} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{cc|c} \boxed{1} & 0 & 1 \\ 0 & \boxed{1} & 2 \end{array}\right]$$

In simple words

A row operation replaces an equation by a combination of equations. Any point that satisfied the old equations satisfies the new ones, and (because the operation can be undone) the other way round. So the **lines rotate** (an *add* operation), **relabel** (a *swap*) or stay exactly the same (a *scale*: $2x - y = 0$ and $4x - 2y = 0$ are the same line), but the **intersection point is never lost and no new point is created**. Gauss–Jordan elimination simply turns the lines until they become $x =$ number (vertical) and $y =$ number (horizontal), and then the answer can be read off.

5.2 Three planes: the same story in space

An equation $ax + by + cz = d$ in three unknowns is a **plane**. The system of Section 1 (Anton No. 5) is three planes through the single point $(3, 1, 2)$. Gauss–Jordan elimination turns them into the three planes $x = 3$, $y = 1$, $z = 2$, which are parallel to the coordinate planes and meet in the same point:



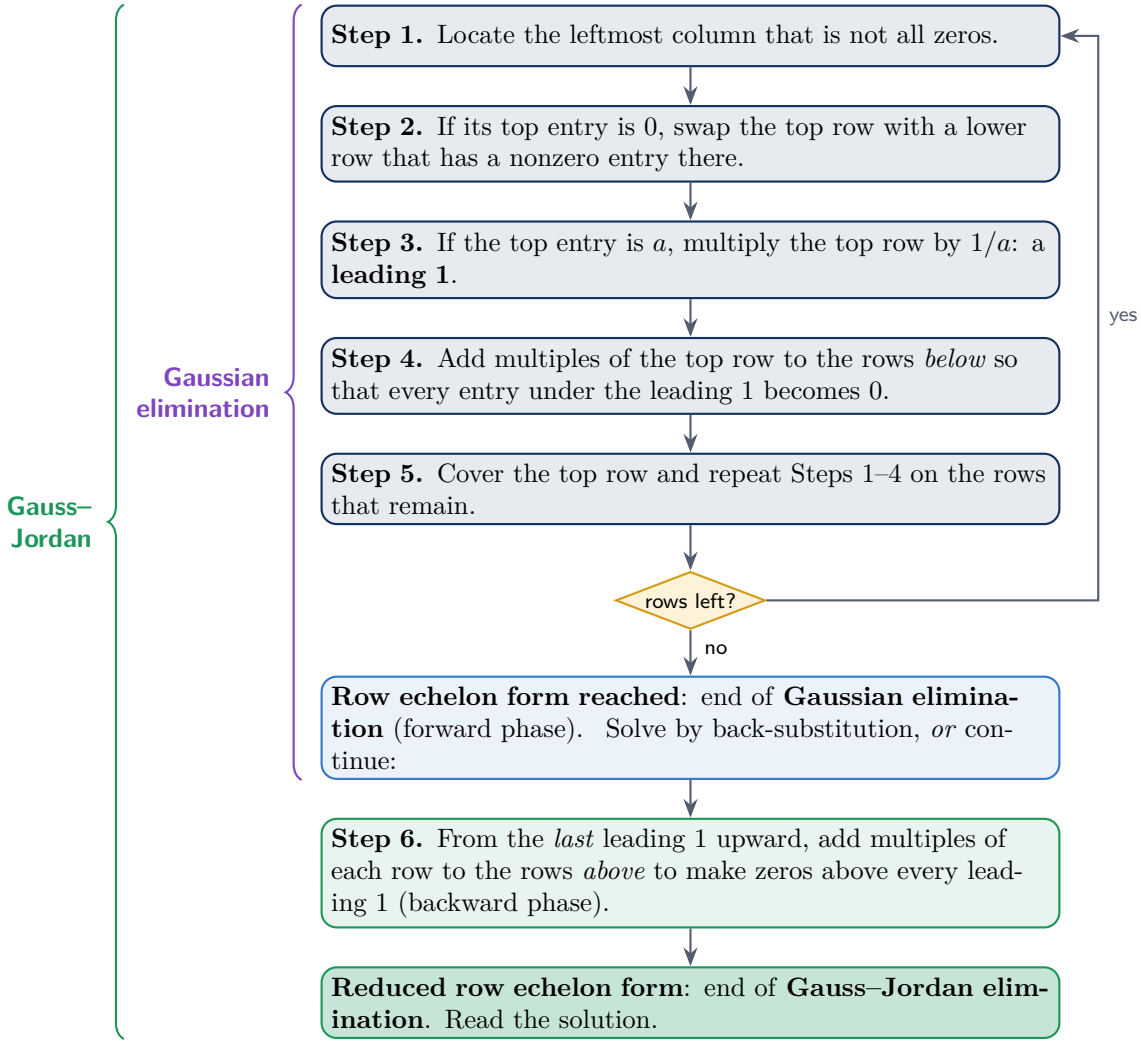
$x_1 + x_2 + 2x_3 = 8$, $-x_1 - 2x_2 + 3x_3 = 1$, $3x_1 - 7x_2 + 4x_3 = 10$ $\longrightarrow$ $x = 3$, $y = 1$, $z = 2$

Why row operations are allowed

Swap: the same equations in another order. **Scale** by $c \neq 0$: c(eq) has exactly the same solutions as (eq), and we can divide by c again. ($c = 0$ is forbidden: it destroys the equation.) **Add:** if **x** solves (eq_i) and (eq_j) it solves (eq_i)+c(eq_j); and subtracting c(eq_j) gives back (eq_i). Systems connected by row operations are called **row equivalent**: they have the **same solution set**.

6 Gaussian and Gauss–Jordan elimination

6.1 The algorithm (Anton's six steps)



Definition 4 (Gaussian / Gauss–Jordan elimination)

Gaussian elimination = the forward phase only (Steps 1–5): it produces a **row echelon form**. **Gauss–Jordan elimination** = forward phase + backward phase (Steps 1–6): it produces the **reduced row echelon form**.

In simple words

Forward phase: go *down* the staircase, cleaning *below* each leading 1. **Backward phase:** climb back *up*, cleaning *above* each leading 1. Gaussian elimination stops halfway and finishes with back-substitution; Gauss–Jordan goes all the way so that the answer is just read.

6.2 Anton's example, one step at a time

[Anton §1.2 · Elimination Methods] Reduce

$$A = \begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}.$$

Step 1. The leftmost nonzero column is column 1. **Step 2.** Its top entry is 0, so swap row 1 with row 2 (which has a nonzero entry, 2):

$$\begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} \mathbf{2} & \mathbf{4} & \mathbf{-10} & \mathbf{6} & \mathbf{12} & \mathbf{28} \\ \mathbf{0} & \mathbf{0} & \mathbf{-2} & \mathbf{0} & \mathbf{7} & \mathbf{12} \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

Step 3. The top entry is $a = 2$; multiply row 1 by $\frac{1}{a} = \frac{1}{2}$ to make a leading 1:

$$\begin{bmatrix} 2 & 4 & -10 & 6 & 12 & 28 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_1} \begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{-5} & \mathbf{3} & \mathbf{6} & \mathbf{14} \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

Step 4. Make the entries below the leading 1 zero (row 2 already has 0; row 3 has 2, so add -2 times row 1):

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_1} \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ \mathbf{0} & \mathbf{0} & \mathbf{5} & \mathbf{0} & \mathbf{-17} & \mathbf{-29} \end{bmatrix}$$

Step 5. Cover row 1 and repeat on rows 2–3. The leftmost nonzero column of the remaining rows is column 3; its top entry is -2, so multiply by $-\frac{1}{2}$; then kill the 5 below it; then cover row 2 and turn the $\frac{1}{2}$ of row 3 into a leading 1:

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{bmatrix} \xrightarrow{R_2 \to -\frac{1}{2}R_2} \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{-\frac{7}{2}} & \mathbf{-6} \\ 0 & 0 & 5 & 0 & -17 & -29 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 5R_2} \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{\frac{1}{2}} & \mathbf{1} \end{bmatrix} \xrightarrow{R_3 \to 2R_3} \begin{bmatrix} \mathbf{(1)} & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & \mathbf{(1)} & 0 & -\frac{7}{2} & -6 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{(1)} & \mathbf{2} \end{bmatrix}$$

This is a **row echelon form**: Gaussian elimination is finished.

Step 6 (backward phase). From the last leading 1 (row 3, column 5) upward: clear the $-\frac{7}{2}$ and the 6 above it, then clear the -5 above the leading 1 of row 2:

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_2 \to R_2 + \frac{7}{2}R_3} \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{1} \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow{R_1 \to R_1 - 6R_3} \begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{-5} & \mathbf{3} & \mathbf{0} & \mathbf{2} \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_1 \to R_1 + 5R_2} \begin{bmatrix} \mathbf{(1)} & \mathbf{2} & \mathbf{0} & \mathbf{3} & \mathbf{0} & \mathbf{7} \\ 0 & 0 & \mathbf{(1)} & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & \mathbf{(1)} & 2 \end{bmatrix}$$

This is the **reduced row echelon form**: Gauss–Jordan elimination is finished. The leading 1's are in columns 1, 3 and 5 (we meet them again as *pivot columns* in Section 10).

6.3 Williams' version: column by column

Williams creates the zeros *above and below* each leading 1 at the same time, moving column by column. The leading entry chosen in each column is called the **pivot**. The result is the same RREF.

Example Gauss–Jordan, column by column [Williams §1.2 · Example 1]

Find the reduced echelon form of

$$\begin{bmatrix} 0 & 0 & 2 & -2 & 2 \\ 3 & 3 & -3 & 9 & 12 \\ 4 & 4 & -2 & 11 & 12 \end{bmatrix}.$$

Column 1: no 0 may be used as a pivot, so swap $R_1 \leftrightarrow R_2$; the pivot 3 becomes a 1 and is used to clear column 1. Column 2 has no nonzero entry below row 1, so it is skipped. Column 3: pivot 2 in row 2, clear above and below. Column 4: pivot in row 3, clear above.

$$\begin{bmatrix} 0 & 0 & 2 & -2 & 2 \\ 3 & 3 & -3 & 9 & 12 \\ 4 & 4 & -2 & 11 & 12 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} \mathbf{3} & \mathbf{3} & \mathbf{-3} & \mathbf{9} & \mathbf{12} \\ \mathbf{0} & \mathbf{0} & \mathbf{2} & \mathbf{-2} & \mathbf{2} \\ 4 & 4 & -2 & 11 & 12 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{3}R_1} \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{-1} & \mathbf{3} & \mathbf{4} \\ 0 & 0 & 2 & -2 & 2 \\ 4 & 4 & -2 & 11 & 12 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 4R_1} \begin{bmatrix} 1 & 1 & -1 & 3 & 4 \\ 0 & 0 & 2 & -2 & 2 \\ \mathbf{0} & \mathbf{0} & \mathbf{2} & \mathbf{-1} & \mathbf{-4} \end{bmatrix} \xrightarrow{R_2 \to \frac{1}{2}R_2} \begin{bmatrix} 1 & 1 & -1 & 3 & 4 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{-1} & \mathbf{1} \\ 0 & 0 & 2 & -1 & -4 \end{bmatrix}$$

$$\xrightarrow[R_3 \to R_3 - 2R_2]{R_1 \to R_1 + R_2} \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{2} & \mathbf{5} \\ 0 & 0 & 1 & -1 & 1 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{-6} \end{bmatrix} \xrightarrow[R_2 \to R_2 + R_3]{R_1 \to R_1 - 2R_3} \begin{bmatrix} \textcircled{1} & \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{17} \\ \mathbf{0} & \mathbf{0} & \textcircled{1} & \mathbf{0} & \mathbf{-5} \\ 0 & 0 & 0 & \textcircled{1} & -6 \end{bmatrix}$$

6.4 Gaussian elimination and back-substitution

Example Solve by Gaussian elimination (video) [Anton · Ex. Set 1.2, No. 5]

Solve $x_1 + x_2 + 2x_3 = 8,\ -x_1 - 2x_2 + 3x_3 = 1,\ 3x_1 - 7x_2 + 4x_3 = 10$.
Augmented matrix. Keep only the numbers: the coefficients of x_1, x_2, x_3 to the left of the bar, the constants to the right.

$$\begin{cases} x_1 + x_2 + 2x_3 = 8 \\ -x_1 - 2x_2 + 3x_3 = 1 \\ 3x_1 - 7x_2 + 4x_3 = 10 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right]$$

Forward phase (Gaussian elimination). Column by column: get a leading 1, then make every entry *below* it zero. Changed rows are printed in **orange**; the leading 1's of the final matrix are circled.

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ \mathbf{0} & \mathbf{-1} & \mathbf{5} & \mathbf{9} \\ \mathbf{0} & \mathbf{-10} & \mathbf{-2} & \mathbf{-14} \end{array}\right]$$

$$\xrightarrow{R_2 \to -R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ \mathbf{0} & \mathbf{1} & \mathbf{-5} & \mathbf{-9} \\ 0 & -10 & -2 & -14 \end{array}\right] \xrightarrow{R_3 \to R_3 + 10R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ \mathbf{0} & \mathbf{0} & \mathbf{-52} & \mathbf{-104} \end{array}\right]$$

$$\xrightarrow{R_3 \to -\frac{1}{52}R_3} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 2 & 8 \\ 0 & \boxed{1} & -5 & -9 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

Read the result. This matrix is in row echelon form. Leading 1's are in columns 1, 2 and 3, so the **leading (pivot) variables** are x_1, x_2 and x_3. No column of unknowns is without a leading 1, so there are **no free variables**: the solution is **unique**. **Back-substitution.** Start from the last nonzero row and move up, each time putting in the values already found:

$$\begin{aligned} &\text{Row 3:} \quad x_3 = 2 \\ &\text{Row 2:} \quad x_2 - 5x_3 = -9 \;\Rightarrow\; x_2 = -9 + 5(2) = 1 \\ &\text{Row 1:} \quad x_1 + x_2 + 2x_3 = 8 \;\Rightarrow\; x_1 = 8 - 1 - 2(2) = 3 \end{aligned}$$

Check (put the answer back into every original equation):

$$\begin{aligned} &\text{Eq. 1:} \quad 3 + 1 + 2(2) = 8 \;\checkmark \\ &\text{Eq. 2:} \quad -3 - 2(1) + 3(2) = 1 \;\checkmark \\ &\text{Eq. 3:} \quad 3(3) - 7(1) + 4(2) = 10 \;\checkmark \end{aligned}$$

Answer. Unique solution: $x_1 = 3, \quad x_2 = 1, \quad x_3 = 2.$

Example The same system by Gauss–Jordan elimination (video) [Anton · Ex. Set 1.2, No. 9]

Same system. The forward phase is exactly the one above; we show it again so that nothing is missing, then add the backward phase.

Augmented matrix. Keep only the numbers: the coefficients of x_1, x_2, x_3 to the left of the bar, the constants to the right.

$$\begin{cases} x_1 + x_2 + 2x_3 = 8 \\ -x_1 - 2x_2 + 3x_3 = 1 \\ 3x_1 - 7x_2 + 4x_3 = 10 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right]$$

Forward phase. Leading 1's and zeros *below* them:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right]$$

$$\xrightarrow{R_2 \to -R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & -10 & -2 & -14 \end{array}\right] \xrightarrow{R_3 \to R_3 + 10R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & -52 & -104 \end{array}\right]$$

$$\xrightarrow{R_3 \to -\frac{1}{52}R_3} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 2 & 8 \\ 0 & \boxed{1} & -5 & -9 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

Backward phase. Starting from the last leading 1 and moving up, make every entry *above* each leading 1 zero:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow[R_2 \to R_2 + 5R_3]{R_1 \to R_1 - 2R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & 3 \\ 0 & \boxed{1} & 0 & 1 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

Read the result. This matrix is in reduced row echelon form. Leading 1's are in columns 1, 2 and 3, so the **leading (pivot) variables** are x_1, x_2 and x_3. No column of unknowns is without a leading 1, so there are **no free variables**: the solution is **unique**. The reduced matrix is the system

$$\begin{cases} x_1 = 3 \\ x_2 = 1 \\ x_3 = 2 \end{cases}$$

Answer. Unique solution: $x_1 = 3, \quad x_2 = 1, \quad x_3 = 2.$

Example From a given REF: pivots and back-substitution (video) [Anton · Ex. Set 1.2, No. 3(a)]

The augmented matrix of a system has been reduced to the row echelon form below. Identify the pivot rows and columns and solve. Unknowns: x_1, x_2, x_3 (columns 1–3); the last column holds the constants.

$$\left[\begin{array}{ccc|c} \textcircled{1} & -3 & 4 & 7 \\ 0 & \textcircled{1} & 2 & 2 \\ 0 & 0 & \textcircled{1} & 5 \end{array}\right]$$

Pivot rows: 1, 2 and 3. **Pivot columns:** 1, 2 and 3. Every column of unknowns contains a leading 1: no free variables, the solution is unique. **Back-substitution** (bottom row first):

Row 3: $x_3 = 5$
Row 2: $x_2 + 2x_3 = 2 \;\Rightarrow\; x_2 = 2 - 2(5) = -8$
Row 1: $x_1 - 3x_2 + 4x_3 = 7 \;\Rightarrow\; x_1 = 7 + 3(-8) - 4(5) = -37$

Check in the REF equations: $-8 + 2(5) = 2$ ✓, $-37 - 3(-8) + 4(5) = -37 + 24 + 20 = 7$ ✓

Answer. $x_1 = -37, \quad x_2 = -8, \quad x_3 = 5.$

	Gaussian elimination	**Gauss–Jordan elimination**
ends with	a row echelon form (not unique)	the reduced row echelon form (unique)
zeros	below the leading 1's	below *and above* the leading 1's
then	back-substitution	read the solution directly
work	less ($\approx 50\%$ fewer operations for large n)	more, but easier by hand
used by	computers (with back-substitution, LU)	hand calculations, theory (rank, basis, …)

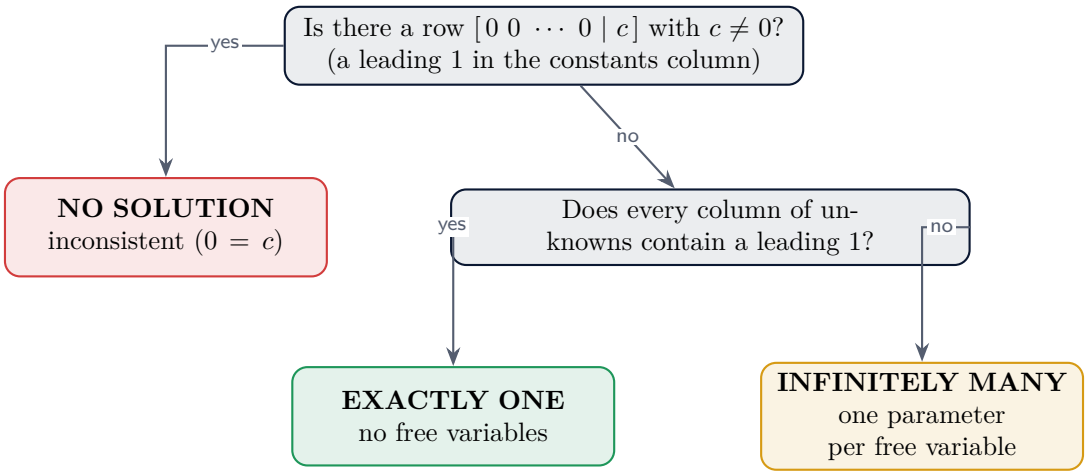
Where is this used?

For a computer, every operation costs time and adds a little round-off error. Williams counts that Gauss–Jordan elimination on n equations in n unknowns needs $\frac{1}{2}n^3 + \frac{1}{2}n^2$ multiplications and $\frac{1}{2}n^3 - \frac{1}{2}n$ additions: for $n = 10$ that is $550 + 495 = 1045$ floating-point operations. Large systems are therefore solved with Gaussian elimination + back-substitution, usually in the LU form of Lecture 5. [Williams §1.2 · Numerical considerations; Anton §1.2 · Roundoff error]

7 One solution, no solution, or infinitely many

7.1 Reading an echelon form

Once the augmented matrix is in echelon form, exactly one of three things happens:



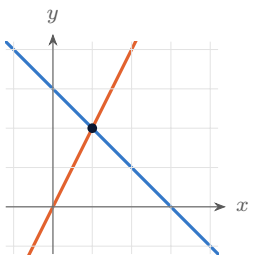
Theorem (Existence and uniqueness, from the echelon form)

Let $[A \mid \mathbf{b}]$ be row reduced to an echelon form. The system is **inconsistent** if and only if the last column contains a leading 1. If it is consistent, it has a **unique** solution when every column of A contains a leading 1, and **infinitely many** solutions otherwise. There is no other possibility (e.g. exactly two solutions is impossible).

Reason. A row $[0 \cdots 0 \mid c]$ is the equation $0 = c$. Otherwise back-substitution always works; each column without a leading 1 is a variable that no equation pins down, and it may take any of infinitely many values.

7.2 In pictures

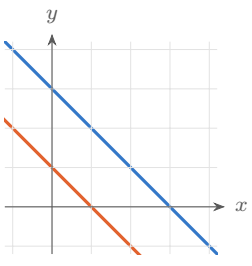
Two equations, two unknowns.



one solution

$x + y = 3,\ 2x - y = 0$

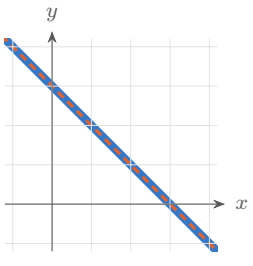
$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array}\right] \to \left[\begin{array}{cc|c} \textcircled{1} & 0 & 1 \\ 0 & \textcircled{1} & 2 \end{array}\right]$$



no solution

$x + y = 3,\ x + y = 1$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 1 & 1 & 1 \end{array}\right] \to \left[\begin{array}{cc|c} \textcircled{1} & 1 & 0 \\ 0 & 0 & \textcircled{1} \end{array}\right]$$



infinitely many

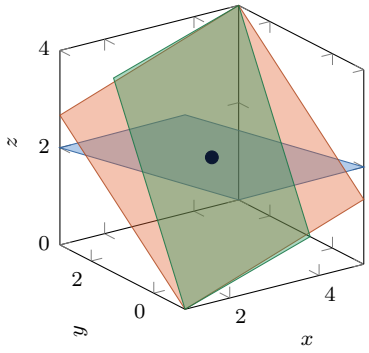
$x + y = 3,\ 2x + 2y = 6$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 2 & 6 \end{array}\right] \to \left[\begin{array}{cc|c} \textcircled{1} & 1 & 3 \\ 0 & 0 & 0 \end{array}\right]$$

intersecting lines | parallel lines: the RREF has $[0\ 0 \mid 1]$ | the same line twice: a zero row and a free variable

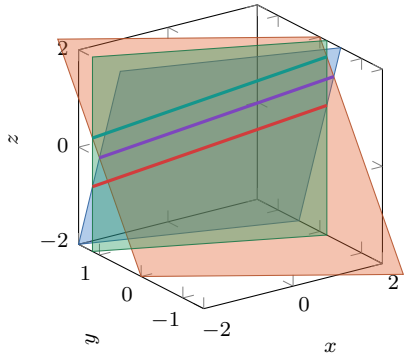
Three equations, three unknowns (planes).

one point: Anton 5



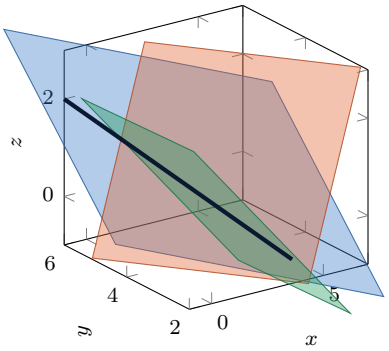
unique solution $(3, 1, 2)$

no point: Williams 5(d)



each pair of planes meets in a line (purple, red, teal), but the three lines are parallel: no common point

a common line: Williams 5(c)



all three planes contain the line $x = 3 - 2t,\ y = 4 + t,\ z = t$

Other "no solution" pictures: two or three parallel planes. Other "infinitely many" pictures: three copies of one plane (a whole plane of solutions).

Example Existence and uniqueness from a REF [Anton §1.2 · Example 8]

These are augmented matrices in x_1, x_2, x_3, x_4, all in row echelon form:

$$\text{(a)}\ \left[\begin{array}{cccc|c} 1 & -3 & 7 & 2 & 5 \\ 0 & 1 & 2 & -4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 0 & \textcircled{1} \end{array}\right] \qquad \text{(b)}\ \left[\begin{array}{cccc|c} 1 & -3 & 7 & 2 & 5 \\ 0 & 1 & 2 & -4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \qquad \text{(c)}\ \left[\begin{array}{cccc|c} 1 & -3 & 7 & 2 & 5 \\ 0 & 1 & 2 & -4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & \textcircled{1} & 0 \end{array}\right]$$

(a) The last row is $0x_1 + 0x_2 + 0x_3 + 0x_4 = 1$: **inconsistent**. (b) The last row is $0 = 0$; leading 1's in columns 1, 2, 3 only, so x_4 is free: **infinitely many**. (c) Leading 1's in all four unknown columns: $x_4 = 0$, then $x_3 = 9 - 6(0) = 9$, then x_2, then x_1: **unique**.

Example Three outcomes from an RREF [Anton §1.2 · Examples 3, 4]

Example 3 (unique). Unknowns: x_1, x_2, x_3, x_4 (columns 1–4); the last column holds the constants.

$$\left[\begin{array}{cccc|c} \textcircled{1} & 0 & 0 & 0 & 3 \\ 0 & \textcircled{1} & 0 & 0 & -1 \\ 0 & 0 & \textcircled{1} & 0 & 0 \\ 0 & 0 & 0 & \textcircled{1} & 5 \end{array}\right]$$

Pivot rows: 1, 2, 3 and 4. **Pivot columns:** 1, 2, 3 and 4. Every column of unknowns contains a leading 1: no free variables, the solution is unique. The matrix is the system

$$\begin{cases} x_1 = 3 \\ x_2 = -1 \\ x_3 = 0 \\ x_4 = 5 \end{cases}$$

Solution: $x_1 = 3, \quad x_2 = -1, \quad x_3 = 0, \quad x_4 = 5.$

Example 4(a). Unknowns: x, y, z (columns 1–3); the last column holds the constants.

$$\left[\begin{array}{ccc|c} \textcircled{1} & 0 & 0 & 0 \\ 0 & \textcircled{1} & 2 & 0 \\ 0 & 0 & 0 & \textcircled{1} \end{array}\right]$$

Pivot rows: 1, 2 and 3. **Pivot columns:** 1, 2 and 4. The leading 1 of row 3 lies in the **constants column**. Row 3 says $0x+0y+0z=1$, which is impossible. **No solution. Solution:** No solution (inconsistent).

Example 4(b). Unknowns: x, y, z (columns 1–3); the last column holds the constants.

$$\left[\begin{array}{ccc|c} \textcircled{1} & 0 & 3 & -1 \\ 0 & \textcircled{1} & -4 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Pivot rows: 1 and 2. **Pivot columns:** 1 and 2. Column 3 of the unknowns contains no leading 1, so z is free: $z=t$. The matrix is the system

$$\begin{cases} x \quad + 3z = -1 \\ \quad y - 4z = 2 \end{cases} \quad \Longrightarrow \quad \begin{array}{l} x = -1 - 3z \\ y = 2 + 4z \end{array}$$

Solution: $x=-1-3t, \quad y=2+4t, \quad z=t \quad (t \text{ arbitrary}).$

Example 4(c). Unknowns: x, y, z (columns 1–3); the last column holds the constants.

$$\left[\begin{array}{ccc|c} \textcircled{1} & -5 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Pivot rows: 1. **Pivot columns:** 1. Columns 2 and 3 of the unknowns contain no leading 1, so y and z are free: $y=s$, $z=t$. The matrix is the system

$$\{x - 5y + z = 4 \quad \Longrightarrow \quad x = 4 + 5y - z$$

Solution: $x=4+5s-t, \quad y=s, \quad z=t \quad (s, t \text{ arbitrary}).$

Example Unique solution with a row swap (video) [Williams · Ex. Set 1.2, No. 5(b)]

Augmented matrix. Keep only the numbers: the coefficients of x_1, x_2, x_3 to the left of the bar, the constants to the right.

$$\begin{cases} x_1 + 2x_2 + 4x_3 = 15 \\ 2x_1 + 4x_2 + 9x_3 = 33 \\ x_1 + 3x_2 + 5x_3 = 20 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 4 & 15 \\ 2 & 4 & 9 & 33 \\ 1 & 3 & 5 & 20 \end{array}\right]$$

Gauss–Jordan elimination. Column by column: make the pivot a leading 1, then create zeros *above and below* it. Changed rows are printed in **orange**; the leading 1's of the final matrix are circled.

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 15 \\ 2 & 4 & 9 & 33 \\ 1 & 3 & 5 & 20 \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 15 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{3} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{5} \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 15 \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{5} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{3} \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{2} & \mathbf{5} \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right] \xrightarrow[R_2 \to R_2 - R_3]{R_1 \to R_1 - 2R_3} \left[\begin{array}{ccc|c} \textcircled{1} & 0 & 0 & \mathbf{-1} \\ \mathbf{0} & \textcircled{1} & \mathbf{0} & \mathbf{2} \\ 0 & 0 & \textcircled{1} & 3 \end{array}\right]$$

Read the result. This matrix is in reduced row echelon form. Leading 1's are in columns 1, 2 and 3, so the **leading (pivot) variables** are x_1, x_2 and x_3. No column of unknowns is without

a leading 1, so there are **no free variables**: the solution is **unique**. The reduced matrix is the system

$$\begin{cases} x_1 = -1 \\ x_2 = 2 \\ x_3 = 3 \end{cases}$$

Check (put the answer back into every original equation):

$$\begin{aligned} &\text{Eq. 1:} \quad (-1) + 2(2) + 4(3) = 15 \ \checkmark \\ &\text{Eq. 2:} \quad 2(-1) + 4(2) + 9(3) = 33 \ \checkmark \\ &\text{Eq. 3:} \quad (-1) + 3(2) + 5(3) = 20 \ \checkmark \end{aligned}$$

Answer. Unique solution: $x_1 = -1, \quad x_2 = 2, \quad x_3 = 3.$

Example No solution (video) [Williams · Ex. Set 1.2, No. 5(d)]

Augmented matrix. Keep only the numbers: the coefficients of x_1, x_2, x_3 to the left of the bar, the constants to the right.

$$\begin{cases} x_1 + 4x_2 + x_3 = 2 \\ x_1 + 2x_2 - x_3 = 0 \\ 2x_1 + 6x_2 = 3 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 1 & 4 & 1 & 2 \\ 1 & 2 & -1 & 0 \\ 2 & 6 & 0 & 3 \end{array}\right]$$

Gauss–Jordan elimination. Column by column: make the pivot a leading 1, then create zeros *above and below* it. Changed rows are printed in **orange**; the leading 1's of the final matrix are circled.

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 2 \\ 1 & 2 & -1 & 0 \\ 2 & 6 & 0 & 3 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 4 & 1 & 2 \\ \mathbf{0} & \mathbf{-2} & \mathbf{-2} & \mathbf{-2} \\ \mathbf{0} & \mathbf{-2} & \mathbf{-2} & \mathbf{-1} \end{array}\right] \xrightarrow{R_2 \to -\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 4 & 1 & 2 \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ 0 & -2 & -2 & -1 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 + 2R_2]{R_1 \to R_1 - 4R_2} \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{-3} & \mathbf{-2} \\ 0 & 1 & 1 & 1 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{array}\right] \xrightarrow[R_2 \to R_2 - R_3]{R_1 \to R_1 + 2R_3} \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{-3} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} \\ 0 & 0 & 0 & \mathbf{1} \end{array}\right]$$

Read the result. Row 3 is $[\,0 \ \cdots \ 0 \mid 1\,]$, i.e. the equation

$$0x_1 + 0x_2 + 0x_3 = 1$$

No numbers can make $0 = 1$, so the system is **inconsistent**.

Answer. No solution (the system is inconsistent).

Example Infinitely many solutions (video) [Williams · Ex. Set 1.2, No. 6(b)]

Augmented matrix. Keep only the numbers: the coefficients of x_1, x_2, x_3 to the left of the bar, the constants to the right.

$$\begin{cases} x_1 + 2x_2 + x_3 = 7 \\ x_1 + 2x_2 + 2x_3 = 11 \\ 2x_1 + 4x_2 + 3x_3 = 18 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 1 & 2 & 2 & 11 \\ 2 & 4 & 3 & 18 \end{array}\right]$$

Gauss–Jordan elimination. Column by column: make the pivot a leading 1, then create zeros *above and below* it. Changed rows are printed in **orange**; the leading 1's of the final matrix are circled.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 1 & 2 & 2 & 11 \\ 2 & 4 & 3 & 18 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{array}\right] \xrightarrow[R_3 \to R_3 - R_2]{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|c} \textcircled{1} & 2 & 0 & 3 \\ 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Read the result. This matrix is in reduced row echelon form. Leading 1's are in columns 1 and 3, so the **leading (pivot) variables** are x_1 and x_3. Row 3 is $0 = 0$: no information, ignore it. Column 2 has *no* leading 1, so x_2 is **free**: $x_2 = t$ (an arbitrary number). Never make a pivot variable free. The reduced matrix is the system

$$\begin{cases} x_1 + 2x_2 = 3 \\ x_3 = 4 \end{cases} \quad \Longrightarrow \quad \begin{aligned} x_1 &= 3 - 2x_2 \\ x_3 &= 4 \end{aligned}$$

Check (put the answer back into every original equation):

$$\begin{aligned} &\text{Eq. 1:} \quad (3 - 2t) + 2(t) + 4 = 7 \ \checkmark \\ &\text{Eq. 2:} \quad (3 - 2t) + 2(t) + 2(4) = 11 \ \checkmark \\ &\text{Eq. 3:} \quad 2(3 - 2t) + 4(t) + 3(4) = 18 \ \checkmark \end{aligned}$$

Answer. Infinitely many solutions: $x_1 = 3 - 2t, \quad x_2 = t, \quad x_3 = 4, \ t$ any real number.

7.3 Free variables: which ones to choose, and which never

Definition 5 (Leading variables, free variables, general solution) [Anton §1.2 · Def. 1; Williams §1.2]

In the echelon form of the augmented matrix, the variables whose columns contain a **leading 1** are the **leading (pivot, basic) variables**; the remaining variables are the **free variables**. Giving the free variables arbitrary values (**parameters** $r, s, t, \ldots$) and solving for the leading variables gives the **general solution**.

x_1	x_2	x_3	x_4	x_5	x_6	
(1)	3	0	4	2	0	0
0	0	(1)	2	0	0	0
0	0	0	0	0	(1)	$\frac{1}{3}$
leading (pivot)	free	leading (pivot)	free	free	leading (pivot)	

Pivot columns 1, 3, 6 → x_1, x_3, x_6 are **determined**.

Non-pivot columns 2, 4, 5 → x_2, x_4, x_5 are **free**: $x_2 = r, \ x_4 = s, \ x_5 = t$.

[Anton §1.2 · Example 5]

This RREF is the system $x_1 + 3x_2 + 4x_4 + 2x_5 = 0$, $x_3 + 2x_4 = 0$, $x_6 = \frac{1}{3}$, so

$$x_1 = -3r - 4s - 2t, \quad x_2 = r, \quad x_3 = -2s, \quad x_4 = s, \quad x_5 = t, \quad x_6 = \tfrac{1}{3}.$$

The rule for free variables (compulsory)

1. **Choose as free exactly the variables of the non-pivot columns** (columns of A with *no* leading 1). Their number is fixed: $\#\text{free} = n - r$, where n = number of unknowns and r = number of leading 1's.
2. **Never make a leading (pivot) variable free.** Each nonzero row has the form $x_{\text{lead}} +$ (terms in variables to its right) $= c$, so it *determines* its leading variable once the variables to its right are known. A leading variable is an *output*, not an input.
3. Give each free variable its own parameter (t; or r, s; or r, s, t) and express every leading variable in terms of them. Do not forget to list the free variables themselves ($x_2 = r$, …) in the answer.

Common mistake

Making a pivot variable free. In Williams No. 6(b) the RREF is

$$\left[\begin{array}{ccc|c} \textcircled{1} & 2 & 0 & 3 \\ 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & 0 \end{array}\right] : \qquad x_1 + 2x_2 = 3, \quad x_3 = 4.$$

Wrong: "let $x_3 = t$ be free". But row 2 says $x_3 = 4$; $t = 0$ would give $x_3 = 0$, which is *not* a solution: $(3, 0, 0)$ gives $x_1 + 2x_2 + 2x_3 = 3 \neq 11$ in the second original equation.
Wrong: "let $x_1 = r$ and $x_2 = s$ both be free". Then $(r, s) = (0, 0)$ gives $(0, 0, 4)$, but $x_1 + 2x_2 = 0 \neq 3$.
Correct: column 2 is the only non-pivot column, so exactly *one* variable, x_2, is free: $x_2 = t$, $x_1 = 3 - 2t$, $x_3 = 4$. ($n - r = 3 - 2 = 1$.)

In simple words

Think of the staircase. Every step (leading 1) is a variable that the equations *solve for*. The columns you jump over between the steps are the variables the equations *cannot* pin down: those are your free choices. Pick them, and everything else is forced.

8 Homogeneous systems

8.1 Trivial and nontrivial solutions

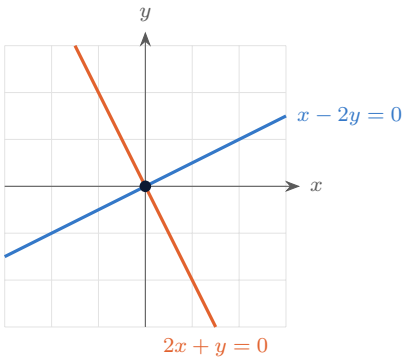
Definition 6 (Homogeneous system) [Anton §1.2; Williams §1.2]

A linear system is **homogeneous** if all constant terms are zero: $A\mathbf{x} = \mathbf{0}$. The solution $x_1 = x_2 = \cdots = x_n = 0$ is the **trivial solution**; any other solution is a **nontrivial solution**.

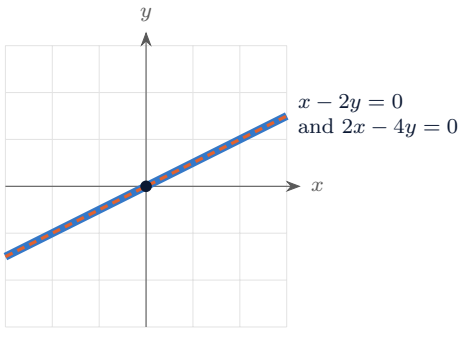
Theorem 1.1 [Williams]

A homogeneous system of linear equations in n variables always has the solution $x_1 = 0, \ldots, x_n = 0$ (the trivial solution).

Reason. Put every variable equal to 0: each equation becomes $0 = 0$. So a homogeneous system is **never inconsistent**, and only two cases remain: **only the trivial solution**, or **infinitely many** solutions (the trivial one and infinitely many nontrivial ones).



only the trivial solution



infinitely many solutions

Homogeneous lines always pass through the origin [Anton §1.2 · Figure 1.2.1]

A column of zeros stays a column of zeros

Every row operation combines zeros with zeros, so the last column of $[A \mid \mathbf{0}]$ remains $\mathbf{0}$ all the time. In practice we may even reduce only A; but in these notes we keep the zero column, so that the method is identical to the general case.

Example A homogeneous system with a free variable [Williams §1.2 · Homogeneous systems]

Augmented matrix. Keep only the numbers: the coefficients of x_1, x_2, x_3, x_4 to the left of the bar, the constants to the right. (The system is *homogeneous*, so the last column is all zeros, and it stays zero under every row operation.)

$$\begin{cases} x_1 + x_2 - x_4 = 0 \\ x_1 + 2x_2 - x_3 + 2x_4 = 0 \\ -x_1 - 2x_2 + 2x_3 - 3x_4 = 0 \end{cases} \longrightarrow \left[\begin{array}{cccc|c} 1 & 1 & 0 & -1 & 0 \\ 1 & 2 & -1 & 2 & 0 \\ -1 & -2 & 2 & -3 & 0 \end{array}\right]$$

Gauss–Jordan elimination. Column by column: make the pivot a leading 1, then create zeros *above and below* it. Changed rows are printed in **orange**; the leading 1's of the final matrix are circled.

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & -1 & 0 \\ 1 & 2 & -1 & 2 & 0 \\ -1 & -2 & 2 & -3 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{cccc|c} 1 & 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & -1 & 2 & -4 & 0 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 + R_2]{R_1 \to R_1 - R_2} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -4 & 0 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right] \xrightarrow[R_2 \to R_2 + R_3]{R_1 \to R_1 - R_3} \left[\begin{array}{cccc|c} \textcircled{1} & 0 & 0 & -3 & 0 \\ 0 & \textcircled{1} & 0 & 2 & 0 \\ 0 & 0 & \textcircled{1} & -1 & 0 \end{array}\right]$$

Read the result. This matrix is in reduced row echelon form. Leading 1's are in columns 1, 2 and 3, so the **leading (pivot) variables** are x_1, x_2 and x_3. Column 4 has *no* leading 1, so x_4 is **free**: $x_4 = t$ (an arbitrary number). Never make a pivot variable free. The reduced matrix is the system

$$\begin{cases} x_1 - 3x_4 = 0 \\ x_2 + 2x_4 = 0 \\ x_3 - x_4 = 0 \end{cases} \Longrightarrow \begin{aligned} x_1 &= 3x_4 \\ x_2 &= -2x_4 \\ x_3 &= x_4 \end{aligned}$$

Check (put the answer back into every original equation):

$$\begin{aligned} &\text{Eq. 1:} \quad (3t) + (-2t) - t = 0 \ \checkmark \\ &\text{Eq. 2:} \quad (3t) + 2(-2t) - t + 2(t) = 0 \ \checkmark \end{aligned}$$

$$\text{Eq. 3:} \quad -(3t) - 2(-2t) + 2(t) - 3(t) = 0 \ \checkmark$$

Answer. Infinitely many solutions: $x_1 = 3t, \quad x_2 = -2t, \quad x_3 = t, \quad x_4 = t,$ t any real number.

8.2 Free variables in homogeneous systems

Theorem 1.2.1 (Free variable theorem for homogeneous systems) [Anton]

If a homogeneous linear system has n unknowns, and the reduced row echelon form of its augmented matrix has r nonzero rows, then the system has $n - r$ free variables.

Reason. Each nonzero row of the RREF has exactly one leading 1, and each leading 1 belongs to one leading variable; the last column is zero, so no leading 1 sits there. Hence there are r leading variables and the other $n - r$ variables are free.

Theorem 1.2.2 [Anton] = **Theorem 1.2** [Williams]

A homogeneous linear system with **more unknowns than equations** has infinitely many solutions (in particular nontrivial ones; one of the solutions is the trivial solution).

Reason. With m equations the RREF has at most m nonzero rows: $r \le m < n$. By Theorem 1.2.1 there are $n - r \ge 1$ free variables, and each free variable can take infinitely many values.

Common mistake

Theorem 1.2.2 is about **homogeneous** systems only. A non-homogeneous system with more unknowns than equations may have **no** solution: $x_1 + x_2 + x_3 = 1, \ x_1 + x_2 + x_3 = 2$. (If such a system *is* consistent, then it has infinitely many solutions.)

Example Nontrivial solutions by inspection (video) [Anton · Ex. Set 1.2, No. 13]

Does the homogeneous system

$$\begin{cases} 2x_1 - 3x_2 + 4x_3 - x_4 = 0 \\ 7x_1 + x_2 - 8x_3 + 9x_4 = 0 \\ 2x_1 + 8x_2 + x_3 - x_4 = 0 \end{cases}$$

have nontrivial solutions? **By inspection:** it is homogeneous with 3 equations and 4 unknowns, so $r \le 3 < 4$ and there is at least one free variable (Theorem 1.2.1); by Theorem 1.2.2 it has infinitely many, hence nontrivial, solutions. No arithmetic is needed.
Confirmation by Gauss–Jordan elimination:
Augmented matrix. Keep only the numbers: the coefficients of x_1, x_2, x_3, x_4 to the left of the bar, the constants to the right. (The system is *homogeneous*, so the last column is all zeros, and it stays zero under every row operation.)

$$\begin{cases} 2x_1 - 3x_2 + 4x_3 - x_4 = 0 \\ 7x_1 + x_2 - 8x_3 + 9x_4 = 0 \\ 2x_1 + 8x_2 + x_3 - x_4 = 0 \end{cases} \longrightarrow \left[\begin{array}{cccc|c} 2 & -3 & 4 & -1 & 0 \\ 7 & 1 & -8 & 9 & 0 \\ 2 & 8 & 1 & -1 & 0 \end{array}\right]$$

Forward phase. Leading 1's and zeros *below* them:

$$\left[\begin{array}{cccc|c} 2 & -3 & 4 & -1 & 0 \\ 7 & 1 & -8 & 9 & 0 \\ 2 & 8 & 1 & -1 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cccc|c} 2 & -3 & 4 & -1 & 0 \\ \mathbf{1} & \mathbf{10} & \mathbf{-20} & \mathbf{12} & \mathbf{0} \\ 2 & 8 & 1 & -1 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cccc|c} \mathbf{1} & \mathbf{10} & \mathbf{-20} & \mathbf{12} & \mathbf{0} \\ \mathbf{2} & \mathbf{-3} & \mathbf{4} & \mathbf{-1} & \mathbf{0} \\ 2 & 8 & 1 & -1 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & 10 & -20 & 12 & 0 \\ \mathbf{0} & \mathbf{-23} & \mathbf{44} & \mathbf{-25} & \mathbf{0} \\ \mathbf{0} & \mathbf{-12} & \mathbf{41} & \mathbf{-25} & \mathbf{0} \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - 2R_3} \left[\begin{array}{cccc|c} 1 & 10 & -20 & 12 & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{-38} & \mathbf{25} & \mathbf{0} \\ 0 & -12 & 41 & -25 & 0 \end{array}\right] \xrightarrow{R_3 \to R_3 + 12R_2} \left[\begin{array}{cccc|c} 1 & 10 & -20 & 12 & 0 \\ 0 & 1 & -38 & 25 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{-415} & \mathbf{275} & \mathbf{0} \end{array}\right]$$

$$\xrightarrow{R_3 \to -\frac{1}{415} R_3} \left[\begin{array}{cccc|c} \mathbf{1} & 10 & -20 & 12 & 0 \\ 0 & \mathbf{1} & -38 & 25 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{-\frac{55}{83}} & \mathbf{0} \end{array}\right]$$

Backward phase. Starting from the last leading 1 and moving up, make every entry *above* each leading 1 zero:

$$\left[\begin{array}{cccc|c} 1 & 10 & -20 & 12 & 0 \\ 0 & 1 & -38 & 25 & 0 \\ 0 & 0 & 1 & -\frac{55}{83} & 0 \end{array}\right] \xrightarrow[R_2 \to R_2 + 38R_3]{R_1 \to R_1 + 20R_3} \left[\begin{array}{cccc|c} \mathbf{1} & \mathbf{10} & \mathbf{0} & \mathbf{-\frac{104}{83}} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{-\frac{15}{83}} & \mathbf{0} \\ 0 & 0 & 1 & -\frac{55}{83} & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 10R_2} \left[\begin{array}{cccc|c} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{\frac{46}{83}} & \mathbf{0} \\ 0 & \mathbf{1} & 0 & -\frac{15}{83} & 0 \\ 0 & 0 & \mathbf{1} & -\frac{55}{83} & 0 \end{array}\right]$$

Read the result. This matrix is in reduced row echelon form. Leading 1's are in columns 1, 2 and 3, so the **leading (pivot) variables** are x_1, x_2 and x_3. Column 4 has *no* leading 1, so x_4 is **free**: $x_4 = t$ (an arbitrary number). Never make a pivot variable free. The reduced matrix is the system

$$\begin{cases} x_1 + \frac{46}{83}x_4 = 0 \\ x_2 - \frac{15}{83}x_4 = 0 \\ x_3 - \frac{55}{83}x_4 = 0 \end{cases} \quad \Longrightarrow \quad \begin{aligned} x_1 &= -\tfrac{46}{83}x_4 \\ x_2 &= \tfrac{15}{83}x_4 \\ x_3 &= \tfrac{55}{83}x_4 \end{aligned}$$

Answer. Infinitely many solutions: $x_1 = -\frac{46}{83}t$, $x_2 = \frac{15}{83}t$, $x_3 = \frac{55}{83}t$, $x_4 = t$, t any real number.

With $t = 83s$: $(x_1, x_2, x_3, x_4) = (-46s,\, 15s,\, 55s,\, 83s)$, e.g. the nontrivial solution $(-46, 15, 55, 83)$. Check eq. 1: $2(-46) - 3(15) + 4(55) - 83 = -92 - 45 + 220 - 83 = 0$ ✓

Example A homogeneous system with two free variables (video) [Williams · Ex. Set 1.2, No. 7(d)]

Augmented matrix. Keep only the numbers: the coefficients of x_1, x_2, x_3, x_4 to the left of the bar, the constants to the right. (The system is *homogeneous*, so the last column is all zeros, and it stays zero under every row operation.)

$$\begin{cases} x_1 + 2x_2 + 4x_4 = 0 \\ -2x_1 - 4x_2 + 3x_3 - 2x_4 = 0 \end{cases} \longrightarrow \left[\begin{array}{cccc|c} 1 & 2 & 0 & 4 & 0 \\ -2 & -4 & 3 & -2 & 0 \end{array}\right]$$

Gauss–Jordan elimination. Column by column: make the pivot a leading 1, then create zeros

above and below it. Changed rows are printed in **orange**; the leading 1's of the final matrix are circled.

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 4 & 0 \\ -2 & -4 & 3 & -2 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + 2R_1} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 4 & 0 \\ 0 & 0 & 3 & 6 & 0 \end{array}\right] \xrightarrow{R_2 \to \frac{1}{3}R_2} \left[\begin{array}{cccc|c} \textcircled{1} & 2 & 0 & 4 & 0 \\ 0 & 0 & \textcircled{1} & 2 & 0 \end{array}\right]$$

Read the result. This matrix is in reduced row echelon form. Leading 1's are in columns 1 and 3, so the **leading (pivot) variables** are x_1 and x_3. Columns 2 and 4 have *no* leading 1, so x_2 and x_4 are **free**: $x_2 = r$, $x_4 = s$ (arbitrary numbers). Never make a pivot variable free. The reduced matrix is the system

$$\begin{cases} x_1 + 2x_2 \quad + 4x_4 = 0 \\ \qquad\quad x_3 + 2x_4 = 0 \end{cases} \quad\Longrightarrow\quad \begin{aligned} x_1 &= -2x_2 - 4x_4 \\ x_3 &= -2x_4 \end{aligned}$$

Check (put the answer back into every original equation):

$$\begin{aligned} \text{Eq. 1:} \quad & (-2r - 4s) + 2(r) + 4(s) = 0 \ \checkmark \\ \text{Eq. 2:} \quad & -2(-2r - 4s) - 4(r) + 3(-2s) - 2(s) = 0 \ \checkmark \end{aligned}$$

Answer. Infinitely many solutions: $x_1 = -2r - 4s$, $x_2 = r$, $x_3 = -2s$, $x_4 = s$, r, s any real numbers.

9 The important theorems in one place

Theorem	Statement	Short reason
Williams Thm 1.1	A homogeneous system always has the trivial solution $\mathbf{x} = \mathbf{0}$.	put all variables $= 0$; every equation becomes $0 = 0$.
Anton Thm 1.2.1 (free variable theorem)	Homogeneous, n unknowns, RREF with r nonzero rows $\Rightarrow$ exactly $n - r$ free variables.	one leading variable per nonzero row; the rest are free.
Anton Thm 1.2.2 = Williams Thm 1.2	Homogeneous with more unknowns than equations $\Rightarrow$ infinitely many solutions.	$r \le m < n$, so $n - r \ge 1$ free variable.
Existence & uniqueness (Anton Ex. 8)	Inconsistent $\Leftrightarrow$ leading 1 in the last column; otherwise unique $\Leftrightarrow$ no free variable.	$0 = c$ is impossible; free variables give infinitely many choices.
Uniqueness of RREF (Anton, Fact 1)	Every matrix has exactly one RREF.	proof: Yuster, *Math. Magazine* 57 (1984).

10 Some facts about echelon forms; pivots

Three facts [Anton §1.2 · Some Facts About Echelon Forms]

1. Every matrix has a **unique** reduced row echelon form: whichever sequence of row operations you use, the same RREF results.
2. Row echelon forms are **not unique**: different sequences of row operations can give different REFs.
3. But all REFs and the RREF of a matrix A have the **same number of zero rows**, and their leading 1's are in the **same positions**.

Definition 7 (Pivot positions, pivot columns, pivot rows, pivots)

The positions of the leading 1's in a (any) row echelon form of A are the **pivot positions** of A. The columns of A that contain them are the **pivot columns**; the rows are the **pivot rows**. A nonzero entry of A itself in a pivot position is a **pivot** of A.

Example Pivot positions and columns [Anton §1.2 · Example 9]

In Section 6 we found a row echelon form of A:

$$A = \begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} \textcircled{1} & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & \textcircled{1} & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 0 & 0 & \textcircled{1} & 2 \end{bmatrix}.$$

The leading 1's are in (row 1, column 1), (row 2, column 3), (row 3, column 5): these are the **pivot positions**. Pivot columns: 1, 3, 5. Pivot rows: 1, 2, 3. Anton marks the **pivots** of A (the nonzero numbers that became the leading 1's) as 2, −10, −5; the first one sits in row 2 of A because Step 2 swapped rows 1 and 2 before the first leading 1 was made.

$$A = \begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ \boxed{2} & 4 & \boxed{-10} & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & \boxed{-5} & -1 \end{bmatrix}$$

shaded: the pivots

pivots: 2, −10, −5

pivot columns

If A is an augmented matrix, the pivot columns identify the leading variables: in Anton's Example 5 the pivot columns are 1, 3, 6, and the leading variables are x_1, x_3, x_6.

Example A matrix with two different row echelon forms (video) [Anton · Ex. Set 1.2, No. 31]

Find two different row echelon forms of $\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$.

First REF:

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} \textcircled{1} & 3 \\ 0 & \textcircled{1} \end{bmatrix}$$

Second REF (start with a swap):

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 2 & 7 \\ 1 & 3 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_1} \begin{bmatrix} 1 & \frac{7}{2} \\ 1 & 3 \end{bmatrix} \xrightarrow{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & \frac{7}{2} \\ 0 & -\frac{1}{2} \end{bmatrix}$$

$$\xrightarrow{R_2 \to -2R_2} \begin{bmatrix} \boxed{1} & \frac{7}{2} \\ 0 & \boxed{1} \end{bmatrix}$$

Both are row echelon forms, and they are different. Their leading 1's are in the same positions $(1,1)$ and $(2,2)$, and both reduce to the same RREF I_2: from the first, $R_1 \to R_1 - 3R_2$; from the second, $R_1 \to R_1 - \frac{7}{2}R_2$.

11 Students' common misconceptions

Each box shows a typical **wrong** piece of work and the **correct** version.

Common mistake

1. "The leading entry of a REF can be any nonzero number."

Wrong: "$\begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}$ is in REF." **Correct** (with the definition of this course, Anton and Williams): property (1) needs leading **1**'s:

$$\begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_1} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \xrightarrow{R_2 \to \frac{1}{3}R_2} \begin{bmatrix} \boxed{1} & 2 \\ 0 & \boxed{1} \end{bmatrix} \text{ is a REF.}$$

(Some other books allow any nonzero leading entry; always follow the definition you are given.)

Common mistake

2. Forgetting the last column.

$$\text{Wrong: } \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 4 & 6 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -2 & 6 \end{array}\right]$$

$$\text{Correct: } \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 4 & 6 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -2 & -9 \end{array}\right]$$

A row operation acts on the *whole* row, bar included: $6 - 3(5) = -9$.

Common mistake

3. Doing two "add" operations at once that use each other.

$$\text{Wrong: } \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \xrightarrow[\substack{R_2 \to R_2 - R_1 \\ \text{(at the same time)}}]{R_1 \to R_1 - R_2} \begin{bmatrix} 0 & -1 \\ 0 & 1 \end{bmatrix}$$

The result has lost information: the original rows are independent, the new ones are not (row 2 = −row 1).

Correct: operations stacked on one arrow must use the *same* unchanged pivot row, e.g. $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - 2R_1$. If an operation needs a row that has just changed, do it on the *next* arrow.

Common mistake

4. Sign slips with negative multipliers.

Wrong: to kill the -2 under a leading 1, "$R_2 \to R_2 - 2R_1$". **Correct:** the multiplier is the

negative of the entry to be killed: entry $-2 \Rightarrow R_2 \to R_2 + 2R_1$, since $-2+2(1)=0$. Always check that the new entry really is 0.

Common mistake

5. "A row operation like $R_2 \to 3R_2 - R_1$ is one step."
It is legal, but it is *two* elementary operations ($R_2 \to 3R_2$, then $R_2 \to R_2 - R_1$). Writing it as one hides the scaling and causes mistakes. And **never** use $R_i \to 0 \cdot R_i$ or $R_i \to R_i - R_i$: that deletes an equation and can create false solutions.

Common mistake

6. "A zero row means infinitely many solutions."
Wrong: Williams No. 8(g) reduces to

$$\left[\begin{array}{cc|c} \boxed{1} & 0 & 3 \\ 0 & \boxed{1} & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right], \quad \text{"so there are infinitely many solutions".}$$

Correct: the zero rows say $0 = 0$ and are simply ignored. Free variables come only from *columns* without a leading 1. Here both columns have one: the solution $(3, -1)$ is **unique**.

Common mistake

7. "The row $[0\ 0\ 0 \mid 0]$ means no solution" / "$[0\ 0\ 0 \mid 5]$ is fine".
Correct: $[0\ 0\ 0 \mid 0]$ is $0 = 0$ (always true, harmless). $[0\ 0\ 0 \mid 5]$ is $0 = 5$ (never true): **no solution**. Only a nonzero number to the right of the bar, with zeros to its left, means inconsistency.

Common mistake

8. Reading a REF as if it were reduced.
Wrong: from

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array}\right] \quad \text{reading "} x_2 = -9\text{".}$$

Correct: row 2 is $x_2 - 5x_3 = -9$, so $x_2 = -9 + 5(2) = 1$. Only in the *reduced* form can a value be read off directly.

Common mistake

9. "The row echelon form of a matrix is unique."
Only the *reduced* form is unique (Anton No. 31):

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \text{ has the REFs } \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & \frac{7}{2} \\ 0 & 1 \end{bmatrix}, \text{ but only one RREF } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Two students can both be right with different REFs, but must get the same RREF.

Common mistake

10. Choosing the wrong free variable. Free variables = non-pivot columns, never a pivot variable, and exactly $n - r$ of them. (See the box in Section 7.)

Common mistake

11. "More unknowns than equations ⇒ infinitely many solutions" / "a homogeneous system can have no solution".
The first is true only for consistent (e.g. homogeneous) systems: $x+y+z=1,\ x+y+z=2$ has none. The second is always false: $\mathbf{x}=\mathbf{0}$ is a solution.

Common mistake

12. Dividing by an expression that might be zero.
Wrong: in Anton No. 25 the last row is $(a^2-16)z=a-4$; "so $z=\dfrac{a-4}{a^2-16}$" for all a. **Correct:** first treat $a^2-16=0$ separately: $a=4$ gives $0=0$ (infinitely many), $a=-4$ gives $0=-8$ (none); only for $a\neq\pm4$ divide (unique).

12 Solved problems on echelon and reduced echelon forms

All textbook exercises of both Exercise Sets 1.2 are solved step by step in the companion PDF *Solved Exercises*. Here is a selection.

Example REF, RREF, both, or neither? [Anton · Ex. Set 1.2, No. 1]

Legend: (1) = a correct leading 1, (2) = a leading entry that breaks a rule, [5] = a nonzero entry in the column of a leading 1 (allowed in REF, *not* in RREF).

	Matrix	Verdict	Reason
(a)	$\begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & \textcircled{1} & 0 \\ 0 & 0 & \textcircled{1} \end{bmatrix}$	**both**	every nonzero row starts with a leading 1; the leading 1's step to the right; each column with a leading 1 has zeros everywhere else.
(b)	$\begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & \textcircled{1} & 0 \\ 0 & 0 & 0 \end{bmatrix}$	**both**	every nonzero row starts with a leading 1; the leading 1's step to the right; the zero rows are at the bottom; each column with a leading 1 has zeros everywhere else.
(c)	$\begin{bmatrix} 0 & \textcircled{1} & 0 \\ 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 \end{bmatrix}$	**both**	every nonzero row starts with a leading 1; the leading 1's step to the right; the zero rows are at the bottom; each column with a leading 1 has zeros everywhere else.
(d)	$\begin{bmatrix} \textcircled{1} & 0 & 3 & 1 \\ 0 & \textcircled{1} & 2 & 4 \end{bmatrix}$	**both**	every nonzero row starts with a leading 1; the leading 1's step to the right; each column with a leading 1 has zeros everywhere else.
(e)	$\begin{bmatrix} \textcircled{1} & 2 & 0 & 3 & 0 \\ 0 & 0 & \textcircled{1} & 1 & 0 \\ 0 & 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$	**both**	every nonzero row starts with a leading 1; the leading 1's step to the right; the zero rows are at the bottom; each column with a leading 1 has zeros everywhere else.

	Matrix	Verdict	Reason
(f)	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$	**both**	A zero matrix: it has no nonzero rows, so conditions 1, 3, 4 say nothing and condition 2 holds (all rows are zero rows).
(g)	$\begin{bmatrix} \textcircled{1} & \boxed{-7} & 5 & 5 \\ 0 & \textcircled{1} & 3 & 2 \end{bmatrix}$	**REF only**	conditions (1)–(3) hold, so it *is* in row echelon form; but **(4)** fails: column 2 contains the leading 1 of row 2 but has the nonzero entry -7 in row 1.

Example REF, RREF, both, or neither? [Anton · Ex. Set 1.2, No. 2]

	Matrix	Verdict	Reason
(a)	$\begin{bmatrix} \textcircled{1} & \boxed{2} & 0 \\ 0 & \textcircled{1} & 0 \\ 0 & 0 & 0 \end{bmatrix}$	**REF only**	conditions (1)–(3) hold, so it *is* in row echelon form; but **(4)** fails: column 2 contains the leading 1 of row 2 but has the nonzero entry 2 in row 1.
(b)	$\begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & \textcircled{1} & 0 \\ 0 & \textcircled{2} & 0 \end{bmatrix}$	**neither**	**(1)** fails: the first nonzero entry of row 3 is 2, not 1; **(3)** fails: the leading entry of row 3 (column 2) is not to the right of the leading entry of row 2 (column 2).
(c)	$\begin{bmatrix} \textcircled{1} & 3 & \boxed{4} \\ 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 \end{bmatrix}$	**REF only**	conditions (1)–(3) hold, so it *is* in row echelon form; but **(4)** fails: column 3 contains the leading 1 of row 2 but has the nonzero entry 4 in row 1.
(d)	$\begin{bmatrix} \textcircled{1} & \boxed{5} & -3 \\ 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 \end{bmatrix}$	**REF only**	conditions (1)–(3) hold, so it *is* in row echelon form; but **(4)** fails: column 2 contains the leading 1 of row 2 but has the nonzero entry 5 in row 1.
(e)	$\begin{bmatrix} \textcircled{1} & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & \textcircled{1} \end{bmatrix}$	**neither**	**(2)** fails: row 2 is a zero row but row 3 below it is not.
(f)	$\begin{bmatrix} \textcircled{1} & 2 & 3 & 4 & 5 \\ \textcircled{1} & 0 & 7 & 1 & 3 \\ 0 & 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$	**neither**	**(3)** fails: the leading entry of row 2 (column 1) is not to the right of the leading entry of row 1 (column 1).
(g)	$\begin{bmatrix} \textcircled{1} & -2 & 0 & 1 \\ 0 & 0 & \textcircled{1} & -2 \end{bmatrix}$	**both**	every nonzero row starts with a leading 1; the leading 1's step to the right; each column with a leading 1 has zeros everywhere else.

Example **Reduced echelon form or not?** [Williams · Ex. Set 1.2, No. 2]

	Matrix	Reduced?	Reason
(a)	$\begin{bmatrix} \textcircled{1} & 0 & 3 & -2 \\ 0 & 0 & \textcircled{1} & 8 \\ 0 & \textcircled{1} & 4 & 9 \end{bmatrix}$	no	**(3)** fails: the leading entry of row 3 (column 2) is not to the right of the leading entry of row 2 (column 3).
(b)	$\begin{bmatrix} \textcircled{1} & 2 & 0 & 0 & 4 \\ 0 & 0 & \textcircled{1} & 0 & 6 \\ 0 & 0 & 0 & \textcircled{1} & 5 \end{bmatrix}$	yes	every nonzero row starts with a leading 1; the leading 1's step to the right; each column with a leading 1 has zeros everywhere else.
(c)	$\begin{bmatrix} \textcircled{1} & 5 & 0 & 2 & 0 \\ 0 & 0 & \textcircled{1} & 9 & 0 \\ 0 & 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$	yes	every nonzero row starts with a leading 1; the leading 1's step to the right; the zero rows are at the bottom; each column with a leading 1 has zeros everywhere else.
(d)	$\begin{bmatrix} \textcircled{1} & 0 & 4 & \boxed{2} & \boxed{6} \\ 0 & \textcircled{1} & 2 & \boxed{3} & \boxed{4} \\ 0 & 0 & 0 & \textcircled{1} & \boxed{2} \\ 0 & 0 & 0 & 0 & \textcircled{1} \end{bmatrix}$	no	conditions (1)–(3) hold, so it *is* in row echelon form; but **(4)** fails: column 4 contains the leading 1 of row 3 but has the nonzero entry 2 in row 1; column 4 contains the leading 1 of row 3 but has the nonzero entry 3 in row 2; column 5 contains the leading 1 of row 4 but has the nonzero entry 6 in row 1; column 5 contains the leading 1 of row 4 but has the nonzero entry 4 in row 2; column 5 contains the leading 1 of row 4 but has the nonzero entry 2 in row 3.
(e)	$\begin{bmatrix} \textcircled{1} & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \textcircled{1} & 2 & 0 & 7 \\ 0 & 0 & 0 & \textcircled{1} & 3 \end{bmatrix}$	no	**(2)** fails: row 2 is a zero row but row 3 below it is not.
(f)	$\begin{bmatrix} \textcircled{1} & 0 & 4 & 0 & 0 \\ 0 & \textcircled{1} & 2 & 0 & 0 \\ 0 & 0 & 0 & \textcircled{1} & 0 \\ 0 & 0 & 0 & 0 & \textcircled{1} \end{bmatrix}$	yes	every nonzero row starts with a leading 1; the leading 1's step to the right; each column with a leading 1 has zeros everywhere else.
(g)	$\begin{bmatrix} \textcircled{1} & 0 & 0 & 5 & 3 \\ 0 & 0 & \textcircled{1} & 0 & 3 \\ 0 & \textcircled{1} & 2 & 3 & 7 \end{bmatrix}$	no	**(3)** fails: the leading entry of row 3 (column 2) is not to the right of the leading entry of row 2 (column 3).
(h)	$\begin{bmatrix} 0 & 0 & \textcircled{1} & 0 & 4 \\ 0 & 0 & 0 & \textcircled{1} & 5 \\ 0 & \textcircled{1} & 0 & 0 & 3 \end{bmatrix}$	no	**(3)** fails: the leading entry of row 3 (column 2) is not to the right of the leading entry of row 2 (column 4).

(i)	$\begin{bmatrix} \boxed{1} & 5 & -3 & 0 & 7 \\ 0 & 0 & 0 & \boxed{1} & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$	yes	every nonzero row starts with a leading 1; the leading 1's step to the right; the zero rows are at the bottom; each column with a leading 1 has zeros everywhere else.

Example **Solution from the reduced echelon form** [Williams · Ex. Set 1.2, No. 3(c), 4(b)]

3(c). Unknowns: x_1, x_2, x_3 (columns 1–3); the last column holds the constants.

$$\left[\begin{array}{ccc|c} \boxed{1} & 3 & 0 & 6 \\ 0 & 0 & \boxed{1} & -2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Pivot rows: 1 and 2. **Pivot columns:** 1 and 3. Column 2 of the unknowns contains no leading 1, so x_2 is free: $x_2 = t$. The matrix is the system

$$\begin{cases} x_1 + 3x_2 = 6 \\ x_3 = -2 \end{cases} \quad \Longrightarrow \quad \begin{aligned} x_1 &= 6 - 3x_2 \\ x_3 &= -2 \end{aligned}$$

Answer: $x_1 = 6 - 3t, \quad x_2 = t, \quad x_3 = -2 \quad (t \text{ arbitrary}).$

4(b). Unknowns: x_1, x_2, x_3, x_4 (columns 1–4); the last column holds the constants.

$$\left[\begin{array}{cccc|c} \boxed{1} & -3 & 2 & 0 & 4 \\ 0 & 0 & 0 & \boxed{1} & -7 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Pivot rows: 1 and 2. **Pivot columns:** 1 and 4. Columns 2 and 3 of the unknowns contain no leading 1, so x_2 and x_3 are free: $x_2 = r$, $x_3 = s$. The matrix is the system

$$\begin{cases} x_1 - 3x_2 + 2x_3 = 4 \\ x_4 = -7 \end{cases} \quad \Longrightarrow \quad \begin{aligned} x_1 &= 4 + 3x_2 - 2x_3 \\ x_4 &= -7 \end{aligned}$$

Answer: $x_1 = 4 + 3r - 2s, \quad x_2 = r, \quad x_3 = s, \quad x_4 = -7 \quad (r, s \text{ arbitrary}).$

Example **Gauss–Jordan elimination, two parameters** [Williams · Ex. Set 1.2, No. 8(c)]

Augmented matrix. Keep only the numbers: the coefficients of x_1, x_2, x_3, x_4 to the left of the bar, the constants to the right.

$$\begin{cases} 2x_1 - 4x_2 + 16x_3 - 14x_4 = 10 \\ -x_1 + 5x_2 - 17x_3 + 19x_4 = -2 \\ x_1 - 3x_2 + 11x_3 - 11x_4 = 4 \\ 3x_1 - 4x_2 + 18x_3 - 13x_4 = 17 \end{cases} \longrightarrow \left[\begin{array}{cccc|c} 2 & -4 & 16 & -14 & 10 \\ -1 & 5 & -17 & 19 & -2 \\ 1 & -3 & 11 & -11 & 4 \\ 3 & -4 & 18 & -13 & 17 \end{array}\right]$$

Gauss–Jordan elimination. Column by column: make the pivot a leading 1, then create zeros *above and below* it. Changed rows are printed in **orange**; the leading 1's of the final matrix are

circled.

$$\left[\begin{array}{cccc|c} 2 & -4 & 16 & -14 & 10 \\ -1 & 5 & -17 & 19 & -2 \\ 1 & -3 & 11 & -11 & 4 \\ 3 & -4 & 18 & -13 & 17 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{cccc|c} \mathbf{1} & \mathbf{-2} & \mathbf{8} & \mathbf{-7} & \mathbf{5} \\ -1 & 5 & -17 & 19 & -2 \\ 1 & -3 & 11 & -11 & 4 \\ 3 & -4 & 18 & -13 & 17 \end{array}\right]$$

$$\xrightarrow[R_4 \to R_4 - 3R_1]{\substack{R_2 \to R_2 + R_1 \\ R_3 \to R_3 - R_1}} \left[\begin{array}{cccc|c} 1 & -2 & 8 & -7 & 5 \\ \mathbf{0} & \mathbf{3} & \mathbf{-9} & \mathbf{12} & \mathbf{3} \\ \mathbf{0} & \mathbf{-1} & \mathbf{3} & \mathbf{-4} & \mathbf{-1} \\ \mathbf{0} & \mathbf{2} & \mathbf{-6} & \mathbf{8} & \mathbf{2} \end{array}\right] \xrightarrow{R_2 \to \frac{1}{3}R_2} \left[\begin{array}{cccc|c} 1 & -2 & 8 & -7 & 5 \\ \mathbf{0} & \mathbf{1} & \mathbf{-3} & \mathbf{4} & \mathbf{1} \\ 0 & -1 & 3 & -4 & -1 \\ 0 & 2 & -6 & 8 & 2 \end{array}\right]$$

$$\xrightarrow[R_4 \to R_4 - 2R_2]{\substack{R_1 \to R_1 + 2R_2 \\ R_3 \to R_3 + R_2}} \left[\begin{array}{cccc|c} \textcircled{1} & \mathbf{0} & \mathbf{2} & \mathbf{1} & \mathbf{7} \\ 0 & \textcircled{1} & -3 & 4 & 1 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{array}\right]$$

Read the result. This matrix is in reduced row echelon form. Leading 1's are in columns 1 and 2, so the **leading (pivot) variables** are x_1 and x_2. Rows 3 and 4 are $0 = 0$: no information, ignore them. Columns 3 and 4 have *no* leading 1, so x_3 and x_4 are **free**: $x_3 = r$, $x_4 = s$ (arbitrary numbers). Never make a pivot variable free. The reduced matrix is the system

$$\begin{cases} x_1 \quad + 2x_3 + \ x_4 = 7 \\ \quad x_2 - 3x_3 + 4x_4 = 1 \end{cases} \implies \begin{array}{l} x_1 = 7 - 2x_3 - x_4 \\ x_2 = 1 + 3x_3 - 4x_4 \end{array}$$

Check (put the answer back into every original equation):

$$\begin{aligned} &\text{Eq. 1:} \quad 2(7 - 2r - s) - 4(1 + 3r - 4s) + 16(r) - 14(s) = 10 \ \checkmark \\ &\text{Eq. 2:} \quad -(7 - 2r - s) + 5(1 + 3r - 4s) - 17(r) + 19(s) = -2 \ \checkmark \\ &\text{Eq. 3:} \quad (7 - 2r - s) - 3(1 + 3r - 4s) + 11(r) - 11(s) = 4 \ \checkmark \\ &\text{Eq. 4:} \quad 3(7 - 2r - s) - 4(1 + 3r - 4s) + 18(r) - 13(s) = 17 \ \checkmark \end{aligned}$$

Answer. Infinitely many solutions: $x_1 = 7 - 2r - s$, $x_2 = 1 + 3r - 4s$, $x_3 = r$, $x_4 = s$, r, s any real numbers.

13 Everything on one page

Augmented matrix: equation ↔ row, variable ↔ column, bar = "=". Three row operations (swap, scale by $c \neq 0$, add a multiple) never change the solution set.

REF: (1) leading 1's, (2) zero rows at the bottom, (3) staircase to the right. **RREF:** also (4) zeros above and below every leading 1. RREF is unique; REF is not.

Gaussian elimination: forward phase → REF, then back-substitution from the bottom row up.

Gauss–Jordan elimination: forward + backward phase → RREF; read the solution directly.

None: a row $[0 \cdots 0 \mid c]$, $c \neq 0$. **One:** leading 1 in every unknown column. **Infinitely many:** otherwise.

Free variables = non-pivot columns, $n - r$ of them; never a pivot variable. Each gets a parameter $r, s, t, \ldots$

Homogeneous $A\mathbf{x} = \mathbf{0}$: always the trivial solution; $n - r$ free variables; $n > m \Rightarrow$ nontrivial solutions.

Pivots: leading-1 positions of any REF = pivot positions; their columns = pivot columns = leading variables.

Five sentences to remember

1. Work with the numbers, not the letters: the augmented matrix.
2. Row operations move the lines and planes but never the solutions.
3. Forward phase gives a REF (Gaussian); backward phase gives the unique RREF (Gauss–Jordan).
4. $[0 \cdots 0 \mid c \neq 0] \Rightarrow$ no solution; a non-pivot column $\Rightarrow$ a free variable $\Rightarrow$ infinitely many.
5. The RREF is the key that opens rank, inverses, independence, bases, null spaces and eigenvectors.

14 Check yourself

Check yourself

1. Is $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$ in RREF? — yes (Williams 1(a))
2. Is $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ in REF? — no: zero row not at the bottom
3. Is $\begin{bmatrix} 1 & 5 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ in RREF? — REF only: the 5 above a leading 1
4. What is the leading entry of $(0, 0, -4, 2)$? — -4 (not a leading 1)
5. An RREF of $[A \mid \mathbf{b}]$ has the row $[0\ 0\ 0 \mid 3]$. Conclusion? — no solution
6. 5 unknowns, 3 leading 1's, consistent. How many parameters? — $5 - 3 = 2$
7. RREF $\left[\begin{array}{ccc|c} 1 & 0 & 3 & -1 \\ 0 & 1 & -4 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$. Solve. — $x = -1 - 3t,\ y = 2 + 4t,\ z = t$ (Anton Ex. 4(b))
8. Which variable is free in $\left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right]$? — x_2 (column 2 has no leading 1)
9. Can a homogeneous system be inconsistent? — never: $\mathbf{x} = \mathbf{0}$ works
10. 4 equations, 6 unknowns, homogeneous. Nontrivial solutions? — yes (Thm 1.2.2)
11. Which multiplier kills the entry -3 under a leading 1 in row 1? — $R_i \to R_i + 3R_1$
12. Find the RREF of $\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$. — I_2

Practice problems

Anton & Rorres, Exercise Set 1.2	
Nos. 1–2	REF, RREF, both, or neither (Sections 3–4)
Nos. 3–4	Pivot rows and columns; solve from a REF / RREF (Sections 6–7, 10)
Nos. 5–8	Gaussian elimination Nos. 9–12: Gauss–Jordan elimination (Section 6)
Nos. 13–22	Homogeneous systems; any method (Sections 8–9)
Nos. 23–30	Consistency with unknown entries; parameters a, b, c (Section 7)
Nos. 31–32, 40, TF	Facts about echelon forms (Section 10)
Williams, Exercise Set 1.2	
Nos. 1–2	Reduced echelon form or not, with a reason
Nos. 3–4	Read the solution from a reduced echelon form
Nos. 5–8	Gauss–Jordan elimination: unique, none, many; homogeneous
Nos. 9–13	Understanding systems of linear equations

All of them are solved step by step in the companion file of solved exercises.